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Remote moving application software and remote moving system

Abstract

When a remote moving control is executed, and a state of controlling a vehicle is a state of changing a shift state of a transmission apparatus of the vehicle, a remote moving application software controls a terminal control unit to cause an operation terminal to generate vibration having a vibration pattern different from the vibration pattern of the vibration generated by the operation terminal when the state of controlling the vehicle is a state other than the state of changing the shift state or output informing sounds having an output pattern different from the output pattern of the informing sounds output by the operation terminal when the state of controlling the vehicle is the state other than the state of changing the shift state.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese patent application No. JP 2021-157689 filed on Sep. 28, 2021, the content of which is hereby incorporated by reference in its entirety.

BACKGROUND

Field
(2) The invention relates to a remote moving application software and a remote moving system.
Description of the Related Art

(3) There is known a remote moving application software and a remote moving system for remotely activating a control unit of an own vehicle to autonomously move and park the own vehicle in a parking space in response to operations applied to a terminal or an operation terminal such as a cell phone (for example, see JP 2015-516772 A).

(4) When the own vehicle is autonomously moved, and an obstacle which will interrupt an autonomous moving of the own vehicle is detected, the known remote moving application software and system inform a user of the operation terminal of the detected obstacle by vibrating the operation terminal. Also, when the known remote moving application software and system autonomously stop the own vehicle or start to autonomously move the own vehicle, the known remote moving application software and system inform the user of the operation terminal that the own vehicle will stop or will start to move by vibrating the operation terminal.

(5) The known remote moving application software and system inform that (i) there is the obstacle, (ii) the own vehicle will stop, and (iii) the own vehicle will start to move by vibrating the operation terminal with different frequencies. Thereby, the user can determine whether (i) there is the obstacle, (ii) the own vehicle will stop, or (iii) the own vehicle will start to move, based on vibration of the operation terminal.

(6) In this regard, the user can easily realize situations that (i) there is the obstacle, (ii) the own vehicle will stop, and (iii) the own vehicle will start to move by seeing the own vehicle, operating the operation terminal. On the other hand, the user cannot easily realize a situation that a shift state of a transmission apparatus has been changed, for example, to K-turn the own vehicle even by seeing the own vehicle. Thus, if the operation terminal can cause the user to easily realize a situation which the user cannot easily realize by seeing the own vehicle, the user can have some benefit. However, the known remote moving application software and system are not configured to use the operation terminal to cause the user to easily realize the situation which the user cannot easily realize with their eyes.

SUMMARY

(7) An object of the invention is to provide a remote moving application software and a remote moving system which can provide the user of the operation terminal with a realization of a situation that the shift state of the transmission apparatus of the vehicle has been changed while the remote moving control to remotely and autonomously move the vehicle is executed.

(8) A remote moving application software according to the invention is installed in a terminal control unit of an operation terminal to enable the terminal control unit to wirelessly communicate a vehicle control unit of a vehicle to cause the vehicle control unit to execute a remote moving control to autonomously move the vehicle in response to operations applied to the operation terminal. Further, the remote moving application software according to the invention is programmed to control the terminal control unit to cause the operation terminal to (i) generate vibration or (ii) output informing sounds, depending on a state of controlling the vehicle when the remote moving control is executed.

(9) When (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is a state of changing a shift state of a transmission apparatus of the vehicle, the remote moving application software according to the invention is programmed to control the terminal control unit to cause the operation terminal to (i) generate the vibration having a vibration pattern different from the vibration pattern of the vibration generated by the operation terminal when the state of controlling the vehicle is a state other than the state of changing the shift state or (ii) output the informing sounds having an output pattern different from the output pattern of the informing sounds output by the operation terminal when the state of controlling the vehicle is the state other than the state of changing the shift state.

(10) According to the invention, while the remote moving control is executed, the vibration is generated in the operation terminal with the different vibration patterns in a shift change situation that the shift state is changed and a situation other than the shift change situation. Alternatively,

while the remote moving control is executed, the informing sounds are output from the operation terminal with the different output patterns in the shift change situation and the situation other than the shift change situation. Thus, the user of the operation terminal can easily realize that the shift state is changed, based on the vibration or the informing sounds.

(11) According to an aspect of the invention, when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is a state of moving the vehicle, the remote moving application software may be programmed to control the terminal control unit to cause the operation terminal to (i) generate the vibration having a predetermined moving vibration pattern or (ii) output the informing sounds having a predetermined moving informing sound output pattern. Further, when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is the state of changing the shift state, the remote moving application software may be programmed to control the terminal control unit to cause the operation terminal to (i) generate the vibration having a shift change vibration pattern different from the predetermined moving vibration pattern or (ii) output the informing sounds having a shift change informing sound output pattern different from the predetermined moving informing sound output pattern.

(12) According to this aspect of the invention, while the remote moving control is executed, the vibration pattern or the output pattern differs between a situation that the vehicle is moved and the situation that the shift state is changed. Thus, the user of the operation terminal can easily determine whether the vehicle is moved, or the shift state is changed, based on the vibration or the informing sounds.

(13) According to another aspect of the invention, when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is a state of braking the vehicle, the remote moving application software may be programmed to control the terminal control unit to cause the operation terminal to (i) generate the vibration having a predetermined braking vibration pattern or (ii) output the informing sounds having a predetermined braking informing sound output pattern. Further, when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is the state of changing the shift state, the remote moving application software may be programmed to control the terminal control unit to cause the operation terminal to (i) generate the vibration having a shift changing vibration pattern different from the predetermined braking vibration pattern or (ii) output the informing sounds having a shift change informing sound output pattern different from the predetermined braking informing sound output pattern.

(14) According to this aspect of the invention, while the remote moving control is executed, the vibration pattern or the output pattern differs between a situation that the vehicle is braked and the situation that the shift state is changed. Thus, the user of the operation terminal can easily determine whether vehicle is braked, or the shift state is changed, based on the vibration or the informing sounds.

(15) According to further another aspect of the invention, when (i) the remote moving control is executed, and (ii) an obstacle is detected around the vehicle, the remote moving application software may be programmed to control the terminal control unit to cause the operation terminal to (i) generate the vibration having the vibration pattern different from the vibration pattern which depends on the state of controlling the vehicle or (ii) output the informing sounds having the output pattern different from the output pattern which depends on the state of controlling the vehicle.

(16) According to this aspect of the invention, when the obstacle is detected while the remote moving control is executed, the vibration is generated in the operation terminal with the vibration pattern which is different from the vibration pattern used to inform the user of the operation terminal of the state of controlling the vehicle. Alternatively, when the obstacle is detected while the remote moving control is executed, the informing sounds are output from the operation terminal with the output pattern which is different from the output pattern used to inform the user of the operation terminal of the state of controlling the vehicle. Thus, the user of the operation terminal can easily realize the state of controlling the vehicle and the obstacle, based on the vibration or the

informing sounds.

(17) Furthermore, a remote moving system according to the invention (i) causes a terminal control unit of an operation terminal and a vehicle control unit of a vehicle to wirelessly communicate each other and (ii) causes the vehicle control unit to execute a remote moving control to autonomously move the vehicle in response to operations applied to the operation terminal. The remote moving system according to the invention is configured to control the terminal control unit to cause the operation terminal to (i) generate vibration or (ii) output informing sounds, depending on a state of controlling the vehicle when the remote moving control is executed.

(18) When (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is a state of changing a shift state of a transmission apparatus of the vehicle, the terminal control unit is configured to cause the operation terminal to (i) generate the vibration having a vibration pattern different from the vibration pattern of the vibration generated by the operation terminal when the state of controlling the vehicle is a state other than the state of changing the shift state or (ii) output the informing sounds having an output pattern different from the output pattern of the informing sounds output by the operation terminal when the state of controlling the vehicle is the state other than the state of changing the shift state.

(19) According to the invention, for the same reason as that described above, the user of the operation terminal can easily realize that the shift state is changed, based on the vibration or the informing sounds.

(20) Elements of the invention are not limited to elements of embodiments and modified examples of the invention described with reference to the drawings. The other objects, features and accompanied advantages of the invention can be easily understood from the embodiments and the modified examples of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view which shows a remote moving system according to an embodiment of the invention.

(2) FIG. 2 is a view which shows a remote moving system according to a modified example of the embodiment of the invention.

(3) FIG. 3 is a view which shows a remote moving system according to another modified example of the embodiment of the invention.

(4) FIG. 4 is a view which shows a terminal control unit and an operation terminal installed with the terminal control unit which configure the remote moving system according to the embodiment of the invention.

(5) FIG. 5 is a view which shows a vehicle control unit and a vehicle (or an own vehicle) installed with the vehicle control unit which configure the remote moving system according to the embodiment of the invention.

(6) FIG. 6 is a view which shows a scene that the own vehicle stops near a parking space.

(7) FIG. 7 is a view which shows the operation terminal which displays an initial screen image on a terminal display.

(8) FIG. 8 is a view which shows the operation terminal which displays a parking-and-pulling-out selection image on the terminal display.

(9) FIG. 9 is a view which shows the operation terminal which displays a parking space selection image on the terminal display.

(10) FIG. 10 is a view which shows a target parking route.

(11) FIG. 11 is a view which shows the operation terminal which displays a parking operation image on the terminal display.

(12) FIG. 12 is a view which shows the operation terminal which displays the parking operation image on the terminal display.

(13) FIG. 13 is a view which shows a scene that the own vehicle is moved forward and rightward by a remote parking control.

(14) FIG. 14 is a view which shows the operation terminal which displays the parking operation image on the terminal display.

(15) FIG. 15 is a view which shows a scene that the own vehicle is moved rearward and leftward by the remote parking control.

(16) FIG. 16 is a view which shows a scene that the own vehicle is moved straight rearward by the remote parking control.

(17) FIG. 17 is a view which shows a scene that parking the own vehicle by the remote parking control is completed.

(18) FIG. 18 is a view which shows the operation terminal which displays the parking operation image on the terminal display.

(19) FIG. 19 is a view which shows a scene that the own vehicle is parked in the parking space.

(20) FIG. 20 is a view which shows the operation terminal which displays a pulling-out direction selection image on the terminal display.

(21) FIG. 21 is a view which shows a target pulling-out route.

(22) FIG. 22 is a view which shows the operation terminal which displays a pulling-out operation image on the terminal display.

(23) FIG. 23 is a view which shows the operation terminal which displays the pulling-out operation image on the terminal display.

(24) FIG. 24 is a view which shows a scene that the own vehicle is moved straight forward by a remote pulling-out control.

(25) FIG. 25 is a view which shows a scene that pulling-out the own vehicle by the remote pulling-out control is completed.

(26) FIG. 26 is a view which shows the operation terminal which displays the pulling-out operation image on the terminal display.

(27) FIG. 27 is a view which shows a time chart which shows vibration and etc. of the operation terminal while a remote moving control is executed.

(28) FIG. 28 is a view which shows a time chart which shows the vibration and etc. of the operation terminal while the remote moving control is executed.

(29) FIG. 29 is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.

(30) FIG. 30 is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.

(31) FIG. 31 is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.

(32) FIG. 32 is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.

(33) FIG. 33 is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.

(34) FIG. 34 is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.

(35) FIG. 35 is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.

(36) FIG. 36 is a view which shows a flowchart of a routine executed by the vehicle control unit according to the embodiment of the invention.

(37) FIG. 37 is a view which shows a flowchart of a routine executed by the vehicle control unit according to the embodiment of the invention.

- (38) FIG. **38** is a view which shows a flowchart of a routine executed by the vehicle control unit according to the embodiment of the invention.
- (39) FIG. **39** is a view which shows a flowchart of a routine executed by the vehicle control unit according to the embodiment of the invention.
- (40) FIG. **40** is a view which shows a flowchart of a routine executed by the vehicle control unit according to the embodiment of the invention.
- (41) FIG. **41** is a view which shows a flowchart of a routine executed by the vehicle control unit according to the embodiment of the invention.
- (42) FIG. **42** is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.
- (43) FIG. **43** is a view which shows a flowchart of a routine executed by the vehicle control unit according to the embodiment of the invention.
- (44) FIG. **44** is a view which shows a flowchart of a routine executed by the terminal control unit according to the embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

- (45) Below, a remote moving application software and a remote moving system according to an embodiment of the invention will be described with reference to the drawings.
- (46) The remote moving application software according to the embodiment of the invention described below is an application software or a remote parking-and-pulling-out application software installed in a terminal control unit (i.e., a control unit of an operation terminal). The remote moving application software causes the terminal control unit and a vehicle control unit (i.e., a control unit of an own vehicle) to wirelessly communicate each other. In addition, the remote moving application software controls the terminal control unit to cause the vehicle control unit to execute a remote moving control or a remote parking-and-pulling-out control to perform an autonomous driving of the vehicle through wireless communication between the terminal control unit and the vehicle control unit. The remote moving control or the remote parking-and-pulling-out control is a control to (i) autonomously move and park the own vehicle in a parking space in response to operations applied to the operation terminal and (ii) autonomously move and pulling out the own vehicle from the parking space in response to the operations applied to the operation terminal.
- (47) The remote moving system described below is a remote parking-and-pulling-out system to cause the vehicle control unit and the terminal control unit to wirelessly communicate each other and control the terminal control unit to cause the vehicle control unit to execute the remote moving control or the remote parking-and-pulling-out control.
- (48) In this regard, the remote moving application software and the remote moving system according to the invention are not limited to the remote parking-and-pulling-out application software and the remote parking-and-pulling-out system described above. The remote moving application software and system according to the invention may be an application software and a system which execute a remote moving control to remotely and autonomously move the own vehicle from the outside of the own vehicle in response to the operations applied to the operation terminal.
- (49) As shown in FIG. **1**, the remote moving system **10** according to the embodiment of the invention includes a terminal control unit **110** (i.e., a control unit installed in an operation terminal **100**) and a vehicle control unit **210** (i.e., a control unit installed on an own vehicle **200**). The remote moving application software according to the embodiment of the invention is installed in the terminal control unit **110**.
- (50) In this embodiment, the terminal control unit **110** and the vehicle control unit **210** of the remote moving system **10** are configured to wirelessly and directly send and receive various signals to and from each other. In this regard, as shown in FIG. **2**, the remote moving system **10** may be configured by (i) the terminal control unit **110**, (ii) the vehicle control unit **210**, and (iii) an internet

300 and may be configured to send and receive signals between the terminal control unit **110** and the vehicle control unit **210** via the internet **300**.

(51) Alternatively, as shown in FIG. **3**, the remote moving system **10** may be configured by (i) the terminal control unit **110**, (ii) the vehicle control unit **210**, (iii) the internet **300**, and (iv) a server **301** provided on the internet **300** and may be configured to send and receive signals between the terminal control unit **110** and **210** via the internet **300** and the server **301**.

(52) <Operation Terminal>

(53) In this embodiment, the operation terminal **100** is a so-called smart phone which is a phone which a person can carry. In this regard, the operation terminal **100** may be any terminal which is separated from the own vehicle **200** and which a user such as a driver DR of the own vehicle **200** can carry to the outside of the own vehicle **200**. For example, the operation terminal **100** may be a so-called smart key which a person can carry or a terminal dedicated to a remote parking-and-pulling-out.

(54) As shown in FIG. **4**, the operation terminal **100** includes the terminal control unit **110**. The terminal control unit **110** includes a terminal ECU **190**.

(55) ECU stands for electronic control unit. The terminal ECU **190** includes a microcomputer as a main component. The terminal ECU **190** includes a CPU (i.e., a terminal CPU **191**), a ROM, a RAM, a non-volatile memory, and an interface. The terminal CPU **191** is configured or programmed to realize various functions by executing instructions, programs, or routines stored in the ROM. In particular, the terminal ECU **190** is installed with the remote moving application software to cause a vehicle ECU **290** (i.e., an ECU of the own vehicle **200**) to execute the remote parking-and-pulling-out control described later in detail.

(56) It should be noted that when the remote moving system **10** is configured as shown in FIG. **3**, the remote moving system **10** may be configured such that a program stored in the server **301** serves a part of functions of the remote moving application software installed in the terminal ECU **190**. Further, when the remote moving system **10** is configured as shown in FIG. **3**, the remote moving system **10** may be configured such that the remote moving application software installed in the terminal ECU **190** is updated by the server **301**.

(57) Further, the operation terminal **100** includes a terminal display **120**, a vibration device **130**, a sound device **140**, and a terminal signal receiving-and-transmitting device **180**.

(58) The terminal display **120** is a device which displays images. Further, a particular physical property of the terminal display **120** changes in response to an object contacting the terminal display **120**. In particular, in this embodiment, the particular physical property of the terminal display **120** changes in response to a finger of a person contacting the terminal display **120**.

(59) The terminal display **120** is electrically connected to the terminal ECU **190**. The terminal ECU **190** can display various images on the terminal display **120**. Further, the terminal ECU **190** detects a change of the particular physical property of the terminal display **120** in response to the object contacting the terminal display **120** and determines a portion of the terminal display **120** which the object contacts, based on the detected change.

(60) It should be noted that the terminal ECU **190** can determine the portion of the terminal display **120** which the object contacts, based on the change of the particular physical property of the terminal display **120** and thus, the terminal display **120** is a contact information providing device which provides information for determining the portion which the object contacts. In this embodiment, the contact information providing device is the terminal display **120**, but is not limited thereto. The contact information providing device may be any device which can provide information to determine the portion of the terminal display **120** which the object contacts. Thus, the contact information providing device may not have a function of displaying images.

(61) The vibration device **130** is a device which generates vibration. The vibration device **130** may be a vibrator. The vibration device **130** is electrically connected to the terminal ECU **190**. The terminal ECU **190** can vibrate the operation terminal **100** by activating the vibration device **130**.

(62) The sound device **140** is a device which outputs informing sounds such as buzzer sounds. The sound device **140** may be a buzzer. The sound device **140** is electrically connected to the terminal ECU **190**. The terminal ECU **190** can output the informing sounds from the sound device **140**.

(63) The terminal signal receiving-and-transmitting device **180** is a device which receives signals wirelessly from the outside of the operation terminal **100** and wirelessly transmits signals to the outside of the operation terminal **100**. The terminal signal receiving-and-transmitting device **180** is electrically connected to the terminal ECU **190**. The terminal ECU **190** can wirelessly transmit various signals to the outside of the operation terminal **100** via the terminal signal receiving-and-transmitting device **180**. In addition, the terminal ECU **190** can receive various wireless signals via the terminal signal receiving-and-transmitting device **180**. In this embodiment, the wireless signals are signals which the vehicle ECU **290** of the own vehicle **200** wirelessly transmits out of the own vehicle **200** via a vehicle receiving-and-transmitting device **280**.

(64) As can be understood, the terminal control unit **110** is configured to wirelessly communicate with the vehicle control unit **210**.

(65) <Own Vehicle>

(66) As shown in FIG. 5, the vehicle control unit **210** is installed on the own vehicle **200**.

(67) The vehicle control unit **210** is a unit which executes a remote parking control and a remote pulling-out control as the remote parking-and-pulling-out control. The remote parking control is a control to park the own vehicle **200** in a designated parking space by autonomously moving the own vehicle **200** and stopping the own vehicle **200** at a point in the designated parking space in response to signals wirelessly transmitted from the operation terminal **100**. The remote pulling-out control is a control to pull out the own vehicle **200** from the designated parking space by autonomously moving the own vehicle **200** and stopping the own vehicle **200** at a point outside of the designated parking space in response to signals wirelessly transmitted from the operation terminal **100**.

(68) As shown in FIG. 5, the vehicle control unit **210** includes a vehicle ECU **290**. The vehicle ECU **290** includes a microcomputer as a main component. The vehicle ECU **290** includes a CPU (i.e., a vehicle CPU **291**), a ROM, a RAM, a non-volatile memory, and an interface. The vehicle CPU **291** is configured or programmed to realize various functions by executing instructions, programs, or routines stored in the ROM. In particular, a program to execute the remote parking-and-pulling-out control is stored in the vehicle ECU **290**.

(69) It should be noted that when the remote moving system **10** is configured as shown in FIG. 3, the remote moving system **10** may be configured such that a part of processes executed by a program of executing the remote parking-and-pulling-out control stored in the vehicle ECU **290** is executed by a program stored in the server **301**. Further, when the remote moving system **10** is configured as shown in FIG. 3, the remote moving system **10** may be configured such that the program of executing the remote parking-and-pulling-out control stored in the vehicle ECU **290** is updated by the server **301**.

(70) <Vehicle Moving Apparatus>

(71) Further, a vehicle moving apparatus **220** is installed on the own vehicle **200**. The vehicle moving apparatus **220** drives, brakes, and steers the own vehicle **200** and performs a shift change of the own vehicle **200**. In this embodiment, the vehicle moving apparatus **220** includes a driving apparatus **221**, a braking apparatus **222**, a steering apparatus **223**, and a transmission apparatus **224**.

(72) <Driving Apparatus>

(73) The driving apparatus **221** outputs a driving force to be applied to the own vehicle **200** to move the own vehicle **200**. The driving apparatus **221** may include an internal combustion engine and/or at least one electric motor. The driving apparatus **221** is electrically connected to the vehicle ECU **290**. The vehicle ECU **290** can control the driving force output from the driving apparatus **221** by controlling operations of the driving apparatus **221**.

(74) <Braking Apparatus>

(75) The braking apparatus **222** outputs a braking force to be applied to the own vehicle **200** to brake the own vehicle **200**. The braking apparatus **222** may include a fluid brake apparatus. The braking apparatus **222** is electrically connected to the vehicle ECU **290**. The vehicle ECU **290** can control the braking force output from the braking apparatus **222** by controlling operations of the braking apparatus **222**.

(76) <Steering Apparatus>

(77) The steering apparatus **223** outputs a steering force to be applied to the own vehicle **200** to steer the own vehicle **200**. The steering apparatus **223** may include a power steering apparatus. The steering apparatus **223** is electrically connected to the vehicle ECU **290**. The vehicle ECU **290** can control the steering force output from the steering apparatus **223** by controlling operations of the steering apparatus **223**.

(78) <Transmission Apparatus>

(79) The transmission apparatus **224** can take any one of (i) a state to transmit the driving force output from the driving apparatus **221** to driven wheels of the own vehicle **200** and (ii) a state to shut off transmitting the driving force output from the driving apparatus **221** to the driven wheels of the own vehicle **200**. In addition, the transmission apparatus **224** can take any one of (i) a state to transmit the driving force to the driven wheels so as to move the own vehicle **200** forward and (ii) a state to transmit the driving force to the driven wheels so as to move the own vehicle **200** rearward. In addition, the transmission apparatus **224** has a function of holding the own vehicle **200** stopped by engaging a pawl part or a parking lock pawl with a gear of the transmission apparatus **224** to lock the gear. Thus, the transmission apparatus **224** functions as a vehicle stop holding apparatus of holding the own vehicle **200** stopped.

(80) The transmission apparatus **224** operates in any one of shift states, i.e., (i) a forward moving driving state SD, (ii) a rearward moving driving state SR, (iii) a neutral state SN, and (iv) a parking state SP. The forward moving driving state SD is a state to transmit the driving force to driven wheels of the own vehicle **200** so as to move the own vehicle **200** forward. The rearward moving driving state SR is a state to transmit the driving force to the driven wheels of the own vehicle **200** so as to move the own vehicle **200** rearward. The neutral state SN is a state to shut off transmitting the driving force to the driven wheels of the own vehicle **200**. The parking state SP is a state to hold the own vehicle **200** stopped.

(81) The transmission apparatus **224** is electrically connected to the vehicle ECU **290**. The vehicle ECU **290** can set the shift state of the transmission apparatus **224** to any one of the forward moving driving state SD, the rearward moving driving state SR, the neutral state SN, and the parking state SP by controlling operations of the transmission apparatus **224**.

(82) <Vehicle Stop Holding Apparatus>

(83) Furthermore, a vehicle stop holding apparatus **230** is installed on the own vehicle **200**. The vehicle stop holding apparatus **230** has a function to hold the own vehicle **200** stopped. In this embodiment, the vehicle stop holding apparatus **230** includes a parking brake apparatus **231**. The parking brake apparatus **231** may be an electric parking brake apparatus or a manual parking brake apparatus. The parking brake apparatus **231** has a function to hold the own vehicle **200** stopped by applying the braking force to wheels of the own vehicle **200**. In particular, the parking brake apparatus **231** has a function to hold the own vehicle **200** stopped by pressing brake pads to brake discs provided on the wheels of the own vehicle **200** to apply the braking force to the wheels of the own vehicle **200**. In this embodiment, the parking brake apparatus **231** is the electric parking brake. Thus, the parking brake apparatus **231** is electrically connected to the vehicle ECU **290**. The vehicle ECU **290** can hold the own vehicle **200** stopped by activating the parking brake apparatus **231**.

(84) <Sensors, Etc.>

(85) Furthermore, an accelerator pedal **241**, an accelerator pedal operation amount sensor **242**, a brake pedal **243**, a brake pedal operation amount sensor **244**, a steering wheel **245**, a steering shaft

246, a steering angle sensor **247**, a steering torque sensor **248**, a remote parking-and-pulling-out request operation device **249**, a vehicle moving speed detecting device **250**, a shift lever **251**, a shift sensor **252**, a surrounding information detecting apparatus **260**, and a vehicle signal receiving-and-transmitting device **280** are installed on the own vehicle **200**.

(86) <Accelerator Pedal Operation Amount Sensor>

(87) The accelerator pedal operation amount sensor **242** detects an operation amount of the accelerator pedal **241**. The accelerator pedal operation amount sensor **242** is electrically connected to the vehicle ECU **290**. The accelerator pedal operation amount sensor **242** sends information on the detected operation amount of the accelerator pedal **241** to the vehicle ECU **290**. The vehicle ECU **290** acquires the operation amount of the accelerator pedal **241** as an accelerator pedal operation amount AP, based on the information sent from the accelerator pedal operation amount sensor **242**.

(88) When the vehicle ECU **290** does not execute the remote parking-and-pulling-out control described later in detail, the vehicle ECU **290** calculates and acquires a required driving force or a required driving torque, based on the accelerator pedal operation amount AP and an own vehicle moving speed (i.e., a moving speed of the own vehicle **200**). The vehicle ECU **290** controls the operations of the driving apparatus **221** so as to output the driving force corresponding to the required driving force. When the vehicle ECU **290** executes the remote parking-and-pulling-out control described later in detail, the vehicle ECU **290** determines a value of the driving force necessary to move the own vehicle **200** as desired by the remote parking-and-pulling-out control and controls the operations of the driving apparatus **221** so as to output the driving force corresponding to the determined value of the driving force.

(89) <Brake Pedal Operation Amount Sensor>

(90) The brake pedal operation amount sensor **244** detects an operation amount of the brake pedal **243**. The brake pedal operation amount sensor **244** is electrically connected to the vehicle ECU **290**. The brake pedal operation amount sensor **244** sends information on the detected operation amount of the brake pedal **243** to the vehicle ECU **290**. The vehicle ECU **290** acquires the operation amount of the brake pedal **243** as a brake pedal operation amount BP, based on the information sent from the brake pedal operation amount sensor **244**.

(91) When the vehicle ECU **290** does not execute the remote parking-and-pulling-out control described later in detail, the vehicle ECU **290** calculates and acquires a required braking force or a required braking torque, based on the brake pedal operation amount BP. The vehicle ECU **290** controls the operations of the braking apparatus **222** so as to output the braking force corresponding to the required braking force. When the vehicle ECU **290** executes the remote parking-and-pulling-out control described later in detail, the vehicle ECU **290** determines a value of the braking force necessary to brake the own vehicle **200** as desired by the remote parking-and-pulling-out control and controls the operations of the braking apparatus **222** so as to output the braking force corresponding to the determined value of the braking force.

(92) <Steering Angle Sensor>

(93) The steering angle sensor **247** detects a rotation angle of the steering shaft **246** with respect to a neutral position thereof. The steering angle sensor **247** is electrically connected to the vehicle ECU **290**. The steering angle sensor **247** sends information on the detected rotation angle of the steering shaft **246** to the vehicle ECU **290**. The vehicle ECU **290** acquires the rotation angle of the steering shaft **246** as a steering angle θ , based on the information sent from the steering angle sensor **247**.

(94) <Steering Torque Sensor>

(95) The steering torque sensor **248** detects a torque which the driver DR of the own vehicle **200** inputs to the steering shaft **246** via the steering wheel **245**. The steering torque sensor **248** is electrically connected to the vehicle ECU **290**. The steering torque sensor **248** sends information on the detected torque to the vehicle ECU **290**. The vehicle ECU **290** acquires the torque which the

driver DR inputs to the steering shaft **246** via the steering wheel **245** as a driver input torque, based on the information sent from the steering torque sensor **248**.

(96) When the vehicle ECU **290** does not execute the remote parking-and-pulling-out control described later in detail, the vehicle ECU **290** acquires a required steering force or a required steering torque, based on the steering angle θ , the driver input torque, and the own vehicle moving speed (i.e., the moving speed of the own vehicle **200**). Then, the vehicle ECU **290** controls the operations of the steering apparatus **223** so as to output the steering torque corresponding to the required steering torque. When the vehicle ECU **290** executes the remote parking-and-pulling-out control described later in detail, the vehicle ECU **290** determines a value of the steering force necessary to brake the own vehicle **200** as desired by the remote parking-and-pulling-out control and controls the operations of the steering apparatus **223** so as to output the steering force corresponding to the determined value of the steering force.

(97) <Remote Parking-And-Pulling-Out Request Operation Device>

(98) The remote parking-and-pulling-out request operation device **249** is a device which is operated by the driver DR to request the vehicle ECU **290** to execute the remote parking-and-pulling-out control described later in detail. The remote parking-and-pulling-out request operation device **249** may be a switch. The remote parking-and-pulling-out request operation device **249** is electrically connected to the vehicle ECU **290**. When the remote parking-and-pulling-out request operation device **249** is operated by the driver DR, the remote parking-and-pulling-out request operation device **249** sends a predetermined signal to the vehicle ECU **290**. The vehicle ECU **290** determines that the remote parking-and-pulling-out control is requested to be executed in response to receiving the predetermined signal from the remote parking-and-pulling-out request operation device **249**.

(99) <Vehicle Moving Speed Detecting Device>

(100) The vehicle moving speed detecting device **250** detects the moving speed of the own vehicle **200**. The vehicle moving speed detecting device **250** may include vehicle wheel rotation speed sensors. The vehicle moving speed detecting device **250** is electrically connected to the vehicle ECU **290**. The vehicle moving speed detecting device **250** sends information on the detected moving speed of the own vehicle **200** to the vehicle ECU **290**. The vehicle ECU **290** acquires the moving speed of the own vehicle **200** as the own vehicle moving speed V , based on the information sent from the vehicle moving speed detecting device **250**.

(101) <Shift Sensor>

(102) The shift sensor **252** detects a set position of the shift lever **251**. The shift lever **251** is a device which is operated by the driver DR of the own vehicle **200**. The set states of the shift lever **251** which the driver DR can set are (i) a forward moving position (or a drive range D), (ii) a rearward moving position (or a rear range R), (iii) a neutral position (or a neutral range N), and (iv) a parking position (or a parking range P). The shift sensor **252** is electrically connected to the vehicle ECU **290**. The shift sensor **252** sends a signal representing the detected set position of the shift lever **251** to the vehicle ECU **290**.

(103) When the shift lever **251** is set in the drive range D, the shift sensor **252** sends a signal representing that the set position of the shift lever **251** is the drive range D to the vehicle ECU **290**. The vehicle ECU **290** controls the shift state of the transmission apparatus **224** to the forward moving driving state SD in response to receiving the signal in question sent from the shift sensor **252**.

(104) When the shift sensor **252** is set in the rear range R, the shift sensor **252** sends a signal representing that the set position of the shift lever **251** is the rear range R to the vehicle ECU **290**. The vehicle ECU **290** controls the shift state of the transmission apparatus **224** to the rearward moving driving state SR in response to receiving the signal in question sent from the shift sensor **252**.

(105) When the shift sensor **252** is set in the neutral range N, the shift sensor **252** sends a signal representing that the set position of the shift lever **251** is the neutral range N to the vehicle ECU

290. The vehicle ECU **290** controls the shift state of the transmission apparatus **224** to the neutral state SN in response to receiving the signal in question sent from the shift sensor **252**.

(106) When the shift sensor **252** is set in the parking range P, the shift sensor **252** sends a signal representing that the set position of the shift lever **251** is the parking range P to the vehicle ECU **290**. The vehicle ECU **290** controls the shift state of the transmission apparatus **224** to the parking state SP in response to receiving the signal in question sent from the shift sensor **252**.

(107) It should be noted that when the vehicle ECU **290** executes the remote parking-and-pulling-out control described later in detail, the vehicle ECU **290** controls the shift state of the transmission apparatus **224** or performs the shift change of the transmission apparatus **224**, depending on a necessity to move the own vehicle **200** as desired by the remote parking-and-pulling-out control.

(108) <Surrounding Information Detecting Apparatus>

(109) The surrounding information detecting apparatus **260** detects information on a situation around the own vehicle **200**. In this embodiment, the surrounding information detecting apparatus **260** includes radio wave sensors **261** and image sensors **262**.

(110) <Radio Wave Sensors>

(111) The radio wave sensor **261** detects information on objects around the own vehicle **200** by using radio waves. The radio wave sensor **261** may be a radar sensor such as a millimeter wave radar, a sound wave sensor such as an ultrasonic sound wave sensor such as a clearance sonar, or an optical sensor such as LiDAR. The radio wave sensor **261** is electrically connected to the vehicle ECU **290**. The radio wave sensor **261** transmits radio waves and receives reflected waves (i.e., the radio waves reflected by objects). The radio wave sensor **261** sends information on the transmitted radio waves and the received radio waves or reflected waves to the vehicle ECU **290**. In other words, the radio wave sensor **261** detects the objects around the own vehicle **200** and send information on the detected objects to the vehicle ECU **290**. The vehicle ECU **290** acquires surrounding detection information IS (i.e., information on the objects around the own vehicle **200**), based on the information (or radio wave information IR or radio wave data) sent from the radio wave sensor **261**. The objects detected by the radio wave sensor **261** may be vehicles, walls, bicycles, and persons.

(112) <Image Sensors>

(113) The image sensor **262** takes images of a view around the own vehicle **200**. The image sensor **262** may be a camera. The image sensor **262** is electrically connected to the vehicle ECU **290**. The image sensor **262** takes the images of the view around the own vehicle **200** and sends information on the taken images to the vehicle ECU **290**.

(114) <Vehicle Signal Receiving-And-Transmitting Device>

(115) The vehicle signal receiving-and-transmitting device **280** has a function of receiving signals wirelessly from the outside of the own vehicle **200** and wirelessly transmitting signals to the outside of the own vehicle **200**. The vehicle signal receiving-and-transmitting device **280** is electrically connected to the vehicle ECU **290**. The vehicle ECU **290** can transmit various signals to the outside of the own vehicle **200** via the vehicle signal receiving-and-transmitting device **280**. In addition, the vehicle ECU **290** can receive various wireless signals via the vehicle signal receiving-and-transmitting device **280**. In this embodiment, the wireless signals are signals which the terminal ECU **190** wirelessly transmits out of the operation terminal **100** via the terminal signal receiving-and-transmitting device **180**.

(116) <Summary of Operations of Remote Moving System>

(117) Next, a summary of operations of the remote moving system **10** will be described.

(118) When the own vehicle **200** is stopped, and the remote parking-and-pulling-out control is requested to be executed, the vehicle control unit **210** becomes a state to receive the wireless signal from the outside thereof and determine whether the received wireless signal is a wireless signal transmitted from a registered operation terminal. The registered operation terminal is an operation terminal which is registered as an operation terminal used for causing the vehicle control unit **210**

to execute the remote parking-and-pulling-out control.

(119) When the driver DR gets out of the own vehicle **200** and applies an application start operation to the terminal display **120** of the operation terminal **100**, the terminal control unit **110** starts the remote moving application software in response to the application start operation and as shown in FIG. 7, displays an initial screen image **G10** on the terminal display **120**.

(120) The application start operation is a touch operation applied to the terminal display **120** for starting the remote moving application software. In this embodiment, the touch operation applied to the terminal display **120** is an operation or a contact operation that a person touches the terminal display **120** with their finger. The initial screen image **G10** includes a remote parking-and-pulling-out start image **G11** and an application end image **G15**. The remote parking-and-pulling-out start image **G11** is an image which displays characters "REMOTE PARKING-AND-PULLING-OUT START." The application end image **G15** is an image which displays characters "END." A remote parking-and-pulling-out start image portion **P11** and an application termination image portion **P15** are portions which accept the touch operation applied by a terminal user UR (i.e., a user of the operation terminal **100**). The remote parking-and-pulling-out start image portion **P11** is a portion of the terminal display **120** which displays the remote parking-and-pulling-out start image **G11**. The application termination image portion **P15** is a portion of the terminal display **120** which displays the application end image **G15**.

(121) When the terminal control unit **110** starts the remote moving application software, the terminal control unit **110** wirelessly transmits an application start signal **S10**. The application start signal **S10** is a signal which represents information on an ID for determining whether the operation terminal **100** is the registered operation terminal, i.e., for identifying the operation terminal **100**.

(122) When the vehicle control unit **210** receives the application start signal **S10**, the vehicle control unit **210** determines whether the operation terminal **100** represented by the application start signal **S10** is the registered operation terminal. The registered operation terminal is the operation terminal which is registered as the operation terminal used for causing the vehicle control unit **210** to execute the remote parking-and-pulling-out control. The operation terminal **100** is the registered operation terminal. Thus, the vehicle control unit **210** determines that the operation terminal **100** is the registered operation terminal when the vehicle control unit **210** receives the application start signal **S10**. In this case, the vehicle control unit **210** executes the remote parking-and-pulling-out control, based on various signals wirelessly transmitted from the operation terminal **100**.

(123) When the vehicle control unit **210** determines that the operation terminal **100** is the registered operation terminal, the vehicle control unit **210** activates vehicle moving equipment such as the surrounding information detecting apparatus **260** and the vehicle moving apparatus **220** or renders itself ready to activate the vehicle moving equipment.

(124) It should be noted that the vehicle control unit **210** may be configured to (i) determine types of assistance to be able to provide, based on the surrounding detection information IS when the vehicle control unit **210** determines that the operation terminal **100** is the registered operation terminal and (ii) wirelessly transmit a result of determining. For example, as described later in detail, the vehicle control unit **210** is configured to execute the remote parking control and the remote pulling-out control. In this regard, the vehicle control unit **210** may be configured to wirelessly transmit a signal representing that only the remote parking control can be executed when the remote pulling-out control cannot be executed and thus, only the remote parking control can be executed. When the terminal control unit **110** receives the signal in question, the terminal control unit **110** adjusts the image displayed on the terminal display **120** such that only the remote parking control can be selected.

(125) Further, the remote moving system **10** described below is a system on which only a control to move the own vehicle **200** rearward and double-park the own vehicle **200** in a designated parking space **31D** can be selected as a control to park the own vehicle **200** in the designated parking space **31D**. In this regard, the remote moving system **10** may be a system on which one of (i) a control to

move the own vehicle **200** forward and double-park the own vehicle **200** in the designated parking space **31D**, (ii) a control to move the own vehicle **200** forward and parallel-park the own vehicle **200** in the designated parking space **31D**, and (iii) a control to move the own vehicle **200** rearward and parallel-park the own vehicle **200** in the designated parking space **31D**, can be selected as the control to park the own vehicle **200** in the designated parking space **31D** in addition to the control to move the own vehicle **200** rearward and double-park the own vehicle **200** in the designated parking space **31D**.

(126) When the touch operation is applied to the remote parking-and-pulling-out start image portion **P11**, the terminal control unit **110** displays a parking-and-pulling-out selection image **G20** on the terminal display **120** as shown in FIG. **8**. In this embodiment, the effective touch operation applied to the remote parking-and-pulling-out start image portion **P11** is a swipe operation. The swipe operation is one of the touch operations which correspond to a slide touch operation, i.e., the touch operation of sliding the finger on the terminal display **120** in one direction.

(127) The parking-and-pulling-out selection image **G20** includes a parking selection image **G21**, a pulling-out selection image **G22**, and an application end image **G25**. The parking selection image **G21** is an image which displays characters "PARKING". The pulling-out selection image **G22** is an image which displays characters "PULLING-OUT". The application end image **G25** is an image which displays characters "END". A parking selection image portion **P21**, a pulling-out selection image portion **P22**, and an application end image portion **P25** are portions which accept the touch operation applied by the terminal user **UR**. The parking selection image portion **P21** is a portion of the terminal display **120** which displays the parking selection image **G21**. The pulling-out selection image portion **P22** is a portion of the terminal display **120** which displays the pulling-out selection image **G22**. The application end image portion **P25** is a portion of the terminal display **120** which displays the application end image **G25**.

(128) <Remote Parking Control>

(129) When the touch operation is applied to the parking selection image portion **P21**, the terminal control unit **110** wirelessly transmits a parking selection signal **S11**. The parking selection signal **S11** is a signal which represents that the touch operation is applied to the parking selection image portion **P21**. In this embodiment, the effective touch operation applied to the parking selection image portion **P21** is a tap operation. The tap operation is one of the touch operations which correspond to an operation that the finger shortly contacts the terminal display **120**.

(130) When the vehicle control unit **210** receives the parking selection signal **S11**, the vehicle control unit **210** detects an available parking space **31C**, i.e., a space in which the own vehicle **200** can be parked, based on the surrounding detection information **IS**. For example, as shown in FIG. **6**, when the own vehicle **200** is stopped near three parking spaces **311** to **313** in which no vehicles are parked, the vehicle control unit **210** detects the parking spaces **311** to **313** as available parking spaces **311C** to **313C**. An example shown in FIG. **6** is an example that there are the parking spaces **311** to **313** at the left side of the own vehicle **200**. In this regard, when there are one or more parking spaces at the right side of the own vehicle **200**, the vehicle control unit **210** detects the one or more parking spaces as the available parking spaces. It should be noted that symbols **400** in the drawings denote lines which defines the parking spaces **31**.

(131) When the vehicle control unit **210** detects the available parking space **31C**, the vehicle control unit **210** acquires a position of the own vehicle **200** with respect to the detected available parking space **31C** and etc.

(132) When the vehicle control unit **210** detects the available parking space **31C** and acquires the position of the own vehicle **200** with respect to the detected available parking space **31C** and etc., the vehicle control unit **210** wirelessly transmits a designated parking space signal **S12** and a vehicle position signal **S13**. The designated parking space signal **S12** is one of signals which represent information on a moving state of the own vehicle **200**. In particular, the designated parking space signal **S12** is a signal which represents information on the available parking space

31C. The vehicle position signal **S13** is one of the signals which represent information on the moving state of the own vehicle **200**. In particular, the vehicle position signal **S13** is a signal which represents information on the position of the own vehicle **200** with respect to the available parking space **31C** and etc.

(133) When the terminal control unit **110** receives the designated parking space signal **S12** and the vehicle position signal **S13**, the terminal control unit **110** causes the terminal display **120** to display a parking space selection image **G30** as shown in FIG. **9**. The parking space selection image **G30** includes an available parking space image **G31**, a vehicle image **G32**, a selection set image **G34**, an application end image **G35**, and a parking-and-pulling-out reselection image **G36**.

(134) The available parking space image **G31** is an image which displays the available parking space **31C**. FIG. **9** shows an example which corresponds to the example shown in FIG. **6**. Thus, the available parking space image **G31** shown in FIG. **9** includes available parking space images **G311** to **G313** which display the available parking spaces **311C** to **313C**. The vehicle image **G32** is an image which displays the own vehicle **200**. The selection set image **G34** is an image which displays characters "SET". The application end image **G35** is an image which displays characters "END". The parking-and-pulling-out reselection image **G36** is an image which displays characters "PARKING-AND-PULLING-OUT RESELECTION".

(135) Further, an available parking space image portion **P31**, a selection set image portion **P34**, an application end image portion **P35**, and a parking-and-pulling-out reselection image portion **P36** are portions which accept the touch operation applied by the terminal user **UR**. The available parking space image portion **P31** is a portion of the terminal display **120** which displays the available parking space image **G31**. FIG. **9** shows the example which corresponds to the example shown in FIG. **6**. Thus, the available parking space image portion **P31** shown in FIG. **9** includes available parking space image portions **P311** to **P313**, i.e., three portions of the terminal display **120** which display the available parking space images **G311** to **G313**, respectively. The selection set image portion **P34** is a portion of the terminal display **120** which displays the selection set image **G34**. The application end image portion **P35** is a portion of the terminal display **120** which displays the application end image **G35**. The parking-and-pulling-out selection image portion **P36** is a portion of the terminal display **120** which displays the parking-and-pulling-out reselection image **G36**.

(136) When the touch operation is applied to the available parking space image portion **P31**, and then the touch operation is applied to the selection set image portion **P34**, the terminal control unit **110** wirelessly transmits a designated parking space signal **S14** or a moving type set signal. In the example shown in FIG. **9**, when the touch operation is applied to any one of the available parking space image portions **P311** to **P313**, and then the touch operation is applied to the selection set image portion **P34**, the terminal control unit **110** wirelessly transmits the designated parking space signal **S14**. The designated parking space signal **S14** is a signal which represents the available parking space **31C** designated as a space in which the own vehicle **200** is parked by the terminal user **UR** applying the touch operation to the available parking space image portion **P31**. In addition, the designated parking space signal **S14** is also a moving type set signal which sets a type of moving of the own vehicle **200** by a vehicle moving process described later in detail.

(137) It should be noted that in this embodiment, the effective touch operation applied to the available parking space image portion **P31** is a long press operation. The long press operation may be called a long tap operation. The long press operation is one of the touch operations that the finger continues to touch the terminal display **120** for a relatively long time.

(138) The parking space selection image **G30** may include a parking direction selection image used to select a parking type of the own vehicle **200** from (i) a parking of the own vehicle **200** in the designated parking space **31D** by moving the own vehicle **200** forward and (ii) a parking of the own vehicle **200** in the designated parking space **31D** by moving the own vehicle **200** rearward. In this case, when the touch operations are applied to the available parking space image portion **P31**

and the parking direction selection image and then, the touch operation is applied to the selection set image portion **P34**, the terminal control unit **110** wirelessly transmits the designated parking space signal **S14**.

(139) When the vehicle control unit **210** receives the designated parking space signal **S14**, the vehicle control unit **210** sets the available parking space **31C** designated by the terminal user UR as a designated parking space **31D**. In the example shown in FIG. **6**, when the vehicle control unit **210** receives the designated parking space signal **S14**, the vehicle control unit **210** sets the available parking space **31C** designated by the terminal user UR, i.e., the available parking space **312C** of the available parking space **311C** to **313C** as the designated parking space **31D**.

(140) When the vehicle control unit **210** sets the designated parking space **31D**, the vehicle control unit **210** starts the remote parking control. When the vehicle control unit **210** starts the remote parking control, the vehicle control unit **210** sets a target parking route R_{in_tgt} as shown in FIG. **10**. The target parking route R_{in_tgt} is a route along which the vehicle control unit **210** moves the own vehicle **200** to park the own vehicle **200** in the designated parking space **31D**. It should be noted that FIG. **10** shows an example in which the available parking space **312C** in the example shown in FIG. **6** is set as the designated parking space **31D**.

(141) When the vehicle control unit **210** sets the target parking route R_{in_tgt} , the vehicle control unit **210** wirelessly transmits a parking remaining distance signal **S15**. The parking remaining distance signal **S15** is one of the signals which represent information on the moving state of the own vehicle **200**. In particular, the parking remaining distance signal **S15** is a signal which represents a parking remaining distance D_{in} , i.e., a distance between the own vehicle **200** and a target parking point P_{in_tgt} . The target parking point P_{in_tgt} is a point in the designated parking space **31D** which the remote parking control causes the own vehicle **200** to reach. The vehicle control unit **210** acquires the parking remaining distance D_{in} , based on the surrounding detection information IS.

(142) On the other hand, when the touch operation is applied to the available parking space image portion **P31**, the terminal control unit **110** causes the terminal display **120** to display a parking operation image **G40** as shown in FIG. **11**. FIG. **11** shows an example which corresponds to the example shown in FIG. **10**. In FIG. **11**, the designated parking space **31D** is the available parking space **31C** or **312C** at the left side of the own vehicle **200**.

(143) The parking operation image **G40** includes a designated parking space image **G41**, a vehicle image **G42**, a parking remaining distance image **G43**, a parking moving operation image **G44**, an application end image **G45**, and a parking-and-pulling-out reselection image **G46**. The designated parking space image **G41** is an image which displays the designated parking space **31D**. The vehicle image **G42** is an image which displays the own vehicle **200**. The parking remaining distance image **G43** is an image which displays the parking remaining distance D_{in} . The parking moving operation image **G44** is an image which defines an area having a predetermined area. The application end image **G45** is an image which displays characters "END". The parking-and-pulling-out reselection image **G46** is an image which displays characters "PARKING-AND-PULLING-OUT RESELECTION". A parking moving operation image portion **P44**, an application end image portion **P45**, and a parking-and-pulling-out reselection image portion **P46** are portions which accept the touch operation applied by the terminal user UR. The parking moving operation image portion **P44** is a portion of the terminal display **120** which displays the parking moving operation image **G44**. The application end image portion **P45** is a portion of the terminal display **120** which displays the application end image **G45**. The parking-and-pulling-out reselection image portion **P46** is a portion of the terminal display **120** which displays the parking-and-pulling-out reselection image **G46**.

(144) When the terminal control unit **110** receives the parking remaining distance signal **S15**, the terminal control unit **110** causes the terminal display **120** to display the parking remaining distance image **G43** which displays the parking remaining distance D_{in} represented by the parking

remaining distance signal **S15**.

(145) While a predetermined operation continues to be applied to the parking moving operation image portion **P44**, the terminal control unit **110** wirelessly transmits a continuous touch operation signal **S16** or a predetermined operation signal. The continuous touch operation signal **S16** is a signal which represents that the predetermined operation is applied to the parking moving operation image portion **P44**. In this embodiment, the predetermined operation applied to the parking moving operation image portion **P44** is the touch operation. In particular, the effective touch operation applied to the parking moving operation image portion **P44** is a continuous touch operation. The continuous touch operation is one of the slide operations. In particular, the continuous touch operation is the slide operation of sliding the finger of the terminal user **UR** on the parking moving operation image portion **P44** as shown by a line **L** in FIG. **12**. Thus, when the continuous touch operation is not carried out, i.e., the finger of the terminal user **UR** touches the parking moving operation image portion **P44**, but stops on the parking moving operation image portion **P44**, the terminal control unit **110** does not transmit the continuous touch operation signal **S16**.

(146) It should be noted that the terminal control unit **110** may be configured to wirelessly transmit a signal which represents a position in the parking moving operation image portion **P44** to which the touch operation is applied in place of the continuous touch operation signal **S16**. In this case, the vehicle control unit **210** receives the signal in question and determines whether the continuous touch operation is applied to the parking moving operation image portion **P44**.

(147) When the vehicle control unit **210** receives the continuous touch operation signal **S16**, the vehicle control unit **210** controls the operations of the transmission apparatus **224** to set the transmission apparatus **224** in the forward moving driving state **SD**. In addition, as shown in FIG. **13**, the vehicle control unit **210** executes a vehicle moving process to control the operations of the vehicle moving apparatus **220** to move the own vehicle **200** forward and rightward along the target parking route **Rin_tgt** and stop the own vehicle **200** when the own vehicle **200** moves a predetermined distance.

(148) While the vehicle control unit **210** receives the continuous touch operation signal **S16**, the vehicle control unit **210** executes the vehicle moving process to move the own vehicle **200** along the target parking route **Rin_tgt**.

(149) Further, the vehicle control unit **210** continues wirelessly transmitting the vehicle position signal **S13**, the parking remaining distance signal **S15**, and a moving direction signal **S17** until the vehicle control unit **210** completes parking the own vehicle **200** in the designated parking space **31D** after the vehicle control unit **210** starts to receive the continuous touch operation signal **S16**. The moving direction signal **S17** is one of the signals which represent the information on the moving state of the own vehicle **200**. In particular, the moving direction signal **S17** is a signal which represents a moving direction of the own vehicle **200**.

(150) When the terminal control unit **110** receives the vehicle position signal **S13**, the terminal control unit **110** causes the terminal display **120** to display the vehicle image **G42** at a portion thereof corresponding to a position of the own vehicle **200** with respect to the designated parking space **31D** which is represented by the vehicle position signal **S13** as shown in FIG. **14**. The vehicle image **G42** shown in FIG. **14** displays the own vehicle **200** which is moved to a position shown in FIG. **13**.

(151) Further, when the terminal control unit **110** receives the parking remaining distance signal **S15**, the terminal control unit **110** causes the terminal display **120** to display the parking remaining distance image **G43** which displays the parking remaining distance **Din** represented by the parking remaining distance signal **S15** as shown in FIG. **14**.

(152) Further, when the terminal control unit **110** receives the moving direction signal **S17**, the terminal control unit **110** causes the terminal display **120** to display the moving direction image **G47** displaying the moving direction of the own vehicle **200** represented by the moving direction signal **S17** at a portion of the terminal display **120** near the vehicle image **G42** as shown in FIG. **14**.

(153) It should be noted that when the terminal user UR puts their finger off the parking moving operation image portion P44, the terminal control unit 110 stops wirelessly transmitting the continuous touch operation signal S16. In this case, the vehicle control unit 210 does not receive the continuous touch operation signal S16. When the vehicle control unit 210 does not receive the continuous touch operation signal S16, the vehicle control unit 210 executes a vehicle stop process to stop the own vehicle 200 by the braking apparatus 222.

(154) When the vehicle control unit 210 moves the own vehicle 200 forward and rightward and stops the own vehicle 200 as shown in FIG. 13, the vehicle control unit 210 executes the vehicle moving process to control the operations of the vehicle moving apparatus 220 as follows. That is, the vehicle control unit 210 changes the shift state of the transmission apparatus 224 from the forward moving driving state SD to the rearward moving driving state SR to K-turn the own vehicle 200. Then, the vehicle control unit 210 moves the own vehicle 200 rearward and leftward, gradually decreasing the steering angle θ such that the steering angle θ becomes zero when a longitudinal direction of the own vehicle 200 becomes parallel to a longitudinal direction of the designated parking space 31D as shown in FIG. 15.

(155) Then, the vehicle control unit 210 executes the vehicle moving process to control the operations of the vehicle moving apparatus 220 to move the own vehicle 200 straight rearward as shown in FIG. 16.

(156) It should be noted that in this embodiment, while the vehicle control unit 210 moves the own vehicle 200 by the remote parking control, the vehicle control unit 210 controls the driving force output from the driving apparatus 221 and the braking force output from the braking apparatus 222 such that the own vehicle moving speed V is maintained at a speed equal to or lower than a predetermined limit vehicle moving speed Vlimit.

(157) Then, when the own vehicle 200 arrives at the target parking point Pin_tgt as shown in FIG. 17, the vehicle control unit 210 stops the own vehicle 200 by the vehicle stop process. Then, the vehicle control unit 210 holds the own vehicle 200 stopped by a vehicle stop holding process, stops the operations of apparatuses such as the surrounding information detecting apparatus 260 and the vehicle moving apparatus 220, and terminates the remote parking control. Thereby, parking the own vehicle 200 in the designated parking space 31D is completed.

(158) It should be noted that in this embodiment, the vehicle stop process is a process to stop the own vehicle 200 by the braking apparatus 222. Further, the vehicle stop holding process is a process to hold the own vehicle 200 stopped by activating the parking brake apparatus 231 or by controlling the operation of the transmission apparatus 224 to put the state of the transmission apparatus 224 in the parking state SP.

(159) Further, when parking the own vehicle 200 is completed, the vehicle control unit 210 wirelessly transmits a parking complete signal S18. The parking complete signal S18 is a signal which represents that parking the own vehicle 200 is completed.

(160) It should be noted that when the vehicle control unit 210 detects an obstacle, based on the surrounding detection information IS while the vehicle control unit 210 moves the own vehicle 200 by the remote parking control, and determines that the own vehicle 200 needs to be stopped, for example, for the reason that the own vehicle 200 is likely to contact the object, the vehicle control unit 210 stops the own vehicle 200 by the vehicle stop process, sets the new target parking route Rin_tgt (or sets the target parking route Rin_tgt again), and starts to move the own vehicle 200.

(161) When the terminal control unit 110 receives the parking complete signal S18, the terminal control unit 110 causes the terminal display 120 to display a parking complete image G49 as shown in FIG. 18. The parking complete image G49 is an image which displays that parking the own vehicle 200 is completed.

(162) <Remote Pulling-Out Control>

(163) When the touch operation is applied to the pulling-out selection image portion P22, the terminal control unit 110 wirelessly transmits a pulling-out selection signal S21. The pulling-out

selection signal **S21** is a signal which represents that the touch operation is applied to the pulling-out selection image portion **P22**. In this embodiment, the effective touch operation applied to the pulling-out selection image portion **P22** is the tap operation.

(164) When the vehicle control unit **210** receives the pulling-out selection signal **S21**, the vehicle control unit **210** acquires the position of the own vehicle **200** with respect to a current parking space **31N** (i.e., the parking space where the own vehicle **200** is parked) and etc., based on the surrounding detection information **IS**. For example, when the own vehicle **200** is parked as shown in FIG. **19**, the parking space **31** is the current parking space **31N**, and the vehicle control unit **210** acquires the position of the own vehicle **200** with respect to the current parking space **31N** and etc.

(165) When the vehicle control unit **210** acquires the position of the own vehicle **200** with respect to the current parking space **31N** and etc., the vehicle control unit **210** wirelessly transmits a vehicle position signal **S23**. The vehicle position signal **S23** is a signal which represents information on the position of the own vehicle **200** with respect to the current parking space **31N** and etc.

(166) When the terminal control unit **110** receives the vehicle position signal **S23**, the terminal control unit **110** causes the terminal display **120** to display a pulling-out direction selection image **G50** as shown in FIG. **20**. The pulling-out direction selection image **G50** includes a current parking space image **G51**, a vehicle image **G52**, a selectable pulling-out direction image **G53**, a selection set image **G54**, an application end image **G55**, and a parking-and-pulling-out reselection image **G56**.

(167) The current parking space image **G51** is an image which displays the current parking space **31N**. The vehicle image **G52** is an image which displays the own vehicle **200**. The selectable pulling-out direction image **G53** is an image which displays pulling-out directions in which the own vehicle **200** can be pulled out. FIG. **20** shows an example which corresponds to the example shown in FIG. **19**. Thus, the selectable pulling-out direction image **G53** shown in FIG. **20** includes a forward direction image **G531** which displays a forward direction and a rearward direction image **G532** which displays a rearward direction. The selection set image **G54** is an image which displays characters “SET”. The application end image **G55** is an image which displays characters “END”. The parking-and-pulling-out reselection image **G56** is an image which displays characters “PARKING-AND-PULLING-OUT RESELECTION”.

(168) Further, a selectable pulling-out direction image portion **P53**, a selection set image portion **P54**, an application end image portion **P55**, and a parking-and-pulling-out reselection image portion **P56** are portions which accept the touch operation applied by the terminal user **UR**. The selectable pulling-out direction image portion **P53** is a portion of the terminal display **120** which displays the selectable pulling-out direction image **G53**. FIG. **20** shows the example which corresponds to the example shown in FIG. **19**. Thus, the selectable pulling-out direction image portion **P53** shown in FIG. **20** includes a forward direction image portion **P531** and a rearward direction image portion **P532**, i.e., two portions of the terminal display **120** which display the forward direction image **G531** and the rearward direction image **G532**, respectively. The selection set image portion **P54** is a portion of the terminal display **120** which displays the selection set image **G54**. The application end image portion **P55** is a portion of the terminal display **120** which displays the application end image **G55**. The parking-and-pulling-out reselection image portion **P56** is a portion of the terminal display **120** which displays the parking-and-pulling-out reselection image **G56**.

(169) When the touch operation is applied to the selectable pulling-out direction image portion **P53**, and then the touch operation is applied to the selection set image portion **P54**, the terminal control unit **110** wirelessly transmits a designated pulling-out direction signal **S22**. In the example shown in FIG. **20**, when the touch operation is applied to any one of the forward direction image portion **P531** and the rearward direction image portion **P532**, and then the touch operation is applied to the selection set image portion **P54**, the terminal control unit **110** wirelessly transmits the

designated pulling-out direction signal S22. The designated pulling-out direction signal S22 is a signal which represents the selectable pulling-out direction designated as a direction in which the own vehicle **200** is pulled out by the terminal user UR applying the touch operation to the selectable pulling-out direction image portion P53. In addition, the designated pulling-out direction signal S22 is also the moving type set signal which sets the type of moving of the own vehicle **200** by the vehicle moving process described later in detail.

(170) It should be noted that in this embodiment, the effective touch operation applied to the selectable pulling-out direction image portion P53 is the long press operation.

(171) When the vehicle control unit **210** receives the designated pulling-out direction signal S22, the vehicle control unit **210** sets the selectable pulling-out direction designated by the terminal user UR as a designated pulling-out direction. In the example shown in FIG. 19, when the vehicle control unit **210** receives the designated pulling-out direction signal S22, the vehicle control unit **210** sets the selectable pulling-out direction designated by the terminal user UR, i.e., the forward direction as the designated pulling-out direction.

(172) When the vehicle control unit **210** sets the designated pulling-out direction, the vehicle control unit **210** starts the remote pulling-out control. When the vehicle control unit **210** starts the remote pulling-out control, the vehicle control unit **210** sets a target pulling-out route Rout_tgt as shown in FIG. 21. The target pulling-out route Rout_tgt is a route along which the vehicle control unit **210** moves the own vehicle **200** in the designated pulling-out direction to pull the own vehicle **200** out of the current parking space 31N. FIG. 21 shows an example in which the forward direction in the example shown in FIG. 19 is set as the designated pulling-out direction.

(173) When the vehicle control unit **210** sets the target pulling-out route Rout_tgt, the vehicle control unit **210** wirelessly transmits a pulling-out remaining distance signal S25. The pulling-out remaining distance signal S25 is a signal which represents a pulling-out remaining distance Dout, i.e., a distance between the own vehicle **200** and a target pulling-out point Pout_tgt. The target pulling-out point Pout_tgt is a point outside of the current parking space 31N which the remote pulling-out control causes the own vehicle **200** to reach. The vehicle control unit **210** acquires the parking-out remaining distance Dout, based on the surrounding detection information IS.

(174) On the other hand, when the touch operation is applied to the selectable pulling-out direction image portion P53, the terminal control unit **110** causes the terminal display **120** to display a pulling-out operation image G60 as shown in FIG. 22. FIG. 22 shows an example which corresponds to the example shown in FIG. 21. In FIG. 22, the designated pulling-out direction is the forward direction.

(175) The pulling-out operation image G60 includes a current parking space image G61, a vehicle image G62, a pulling-out remaining distance image G63, a pulling-out moving operation image G64, an application end image G65, a parking-and-pulling-out reselection image G66, and a pulling-out direction image G67. The current parking space image G61 is an image which displays the current parking space 31N. The vehicle image G62 is an image which displays the own vehicle **200**. The pulling-out remaining distance image G63 is an image which displays the parking-out remaining distance Dout. The pulling-out moving operation image G64 is an image which defines an area having a predetermined area. The application end image G65 is an image which displays characters "END". The parking-and-pulling-out reselection image G66 is an image which displays characters "PARKING-AND-PULLING-OUT RESELECTION". The pulling-out direction image G67 is an image which displays the designated pulling-out direction.

(176) A pulling-out moving operation image portion P64, an application end image portion P65, and a parking-and-pulling-out reselection image portion P66 are portions which accept the touch operation applied by the terminal user UR. The pulling-out moving operation image portion P64 is a portion of the terminal display **120** which displays the pulling-out moving operation image G64. The application end image portion P65 is a portion of the terminal display **120** which displays the application end image G65. The parking-and-pulling-out reselection image portion P66 is a portion

of the terminal display **120** which displays the parking-and-pulling-out reselection image **G66**.

(177) When the terminal control unit **110** receives the pulling-out remaining distance signal **S25**, the terminal control unit **110** causes the terminal display **120** to display the pulling-out remaining distance image **G63** which displays the parking-out remaining distance **Dout** represented by the pulling-out remaining distance signal **S25**.

(178) While the touch operation continues to be applied to the pulling-out moving operation image portion **P64**, the terminal control unit **110** wirelessly transmits a continuous touch operation signal **S26** or a predetermined operation signal. The continuous touch operation signal **S26** is a signal which represents that the predetermined operation is applied to the pulling-out moving operation image portion **P64**. In this embodiment, the predetermined operation applied to the pulling-out moving operation image portion **P64** is the touch operation. In particular, the effective touch operation applied to the pulling-out moving operation image portion **P64** is the continuous touch operation. The continuous touch operation is one of the slide operations. In particular, the continuous touch operation is the slide operation of sliding the finger of the terminal user **UR** on the pulling-out moving operation image portion **P64** as shown by a line **L** in FIG. 23. Thus, when the continuous touch operation is not carried out, i.e., the finger of the terminal user **UR** touches the pulling-out moving operation image portion **P64**, but stops on the pulling-out moving operation image portion **P64**, the terminal control unit **110** does not transmit the continuous touch operation signal **S26**.

(179) It should be noted that the terminal control unit **110** may be configured to wirelessly transmit a signal which represents a position in the pulling-out moving operation image portion **P64** to which the touch operation is applied in place of the continuous touch operation signal **S26**. In this case, the vehicle control unit **210** receives the signal in question and determines whether the continuous touch operation is applied to the pulling-out moving operation image portion **P64**.

(180) When the vehicle control unit **210** receives the continuous touch operation signal **S26**, the vehicle control unit **210** controls the operations of the transmission apparatus **224** to set the transmission apparatus **224** in the forward moving driving state **SD**. In addition, as shown in FIG. 24, the vehicle control unit **210** executes the vehicle moving process to control the operations of the vehicle moving apparatus **220** to move the own vehicle **200** forward along the target pulling-out route **Rout_tgt** and stop the own vehicle **200** when the own vehicle **200** arrives at the target pulling-out point **Pout_tgt**.

(181) While the vehicle control unit **210** receives the continuous touch operation signal **S26**, the vehicle control unit **210** executes the vehicle moving process to move the own vehicle **200** along the target pulling-out route **Rout_tgt**.

(182) Further, the vehicle control unit **210** continues wirelessly transmitting the vehicle position signal **S23** and the pulling-out remaining distance signal **S25** until the vehicle control unit **210** completes pulling the own vehicle **200** out of the current parking space **31N** after the vehicle control unit **210** starts to receive the continuous touch operation signal **S26**.

(183) When the terminal control unit **110** receives the vehicle position signal **S23**, the terminal control unit **110** causes the terminal display **120** to display the vehicle image **G62** at a portion thereof corresponding to a position of the own vehicle **200** with respect to the current parking space **31N** which is represented by the vehicle position signal **S23** as shown in FIG. 23. The vehicle image **G62** shown in FIG. 23 displays the own vehicle **200** which is moved to a position shown in FIG. 24.

(184) Further, when the terminal control unit **110** receives the pulling-out remaining distance signal **S25**, the terminal control unit **110** causes the terminal display **120** to display the pulling-out remaining distance image **G63** which displays the parking-out remaining distance **Dout** represented by the pulling-out remaining distance signal **S25** as shown in FIG. 23.

(185) It should be noted that when the terminal user **UR** puts their finger off the pulling-out moving operation image portion **P64**, the terminal control unit **110** stops wirelessly transmitting the

continuous touch operation signal S26. In this case, the vehicle control unit 210 does not receive the continuous touch operation signal S26. When the vehicle control unit 210 does not receive the continuous touch operation signal S26, the vehicle control unit 210 executes the vehicle stop process to stop the own vehicle 200 by the braking apparatus 222.

(186) It should be noted that in this embodiment, while the vehicle control unit 210 moves the own vehicle 200 by the remote pulling-out control, the vehicle control unit 210 controls the driving force output from the driving apparatus 221 and the braking force output from the braking apparatus 222 such that the own vehicle moving speed V is maintained at a speed equal to or lower than the predetermined limit vehicle moving speed V_{limit} .

(187) Then, when the own vehicle 200 arrives at the target pulling-out point P_{out_tgt} as shown in FIG. 25, the vehicle control unit 210 stops the own vehicle 200 by the vehicle stop process. Then, the vehicle control unit 210 holds the own vehicle 200 stopped by the vehicle stop holding process, stops the operations of apparatuses such as the surrounding information detecting apparatus 260 and the vehicle moving apparatus 220, and terminates the remote pulling-out control. Thereby, pulling-out the own vehicle 200 of the current parking space 31N is completed.

(188) Further, when pulling out the own vehicle 200 is completed, the vehicle control unit 210 wirelessly transmits a pulling-out complete signal S28. The pulling-out complete signal S28 is a signal which represents that pulling out the own vehicle 200 is completed.

(189) When the terminal control unit 110 receives the pulling-out complete signal S28, the terminal control unit 110 causes the terminal display 120 to display a pulling-out complete image G69 as shown in FIG. 26. The pulling-out complete image G69 is an image which displays that pulling out the own vehicle 200 is completed.

(190) It should be noted that when the vehicle control unit 210 determines to need to K-turn the own vehicle 200 to pulling out the own vehicle 200 while the vehicle control unit 210 executes the remote pulling-out control, the vehicle control unit 210 changes the shift state of the transmission apparatus 224 and K-turns the own vehicle 200.

(191) <Termination of Application Software, and Etc.>

(192) When the touch operation is applied to any one of the application termination image portions P15 to P65 (i.e., the application termination image portions P15, P25, P35, P45, P55, and P65), the terminal control unit 110 terminates the remote moving application software and wirelessly transmits a control termination command signal S30. The control termination command signal S30 is a signal which commands terminating the remote parking-and-pulling-out control.

(193) When the touch operation is applied to any one of the parking-and-pulling-out reselection image portions P36 to P66 (i.e., the parking-and-pulling-out reselection image portions P36, P46, P56, and P66), the terminal control unit 110 causes the terminal display 120 to display the parking-and-pulling-out selection image G20 and wirelessly transmits the control termination command signal S30.

(194) When the vehicle control unit 210 receives the control termination command signal S30, the vehicle control unit 210 stops the own vehicle 200 by the vehicle stop process. In addition, the vehicle control unit 210 holds the own vehicle 200 stopped by the vehicle stop holding process, stops the operations of apparatuses such as the surrounding information detecting apparatus 260 and the vehicle moving apparatus 220, and terminates the remote parking control or the remote pulling-out control.

(195) It should be noted that the terminal control unit 110 may be configured to cause the terminal display 120 to display an image which defines a portion of the terminal display 120 to which the touch operation to return a screen of the terminal display 120 to the previous screen is applied when the remote moving application software is programmed to change the screen of the terminal display 120 many times while the remote moving application software is executed.

(196) <Vibration Process>

(197) As described above, the terminal user UR continuously applies the touch operation with their

finger to the terminal display **120** of the operation terminal **100** when the own vehicle **200** is autonomously moved by the remote moving control or the remote parking-and-pulling-out control. For ensuring a moving safety of the own vehicle **200**, the terminal user UR is required to apply the touch operation to the terminal display **120**, watching the moving state of the own vehicle **200** without watching the terminal display **120**.

(198) If information on a state of controlling the own vehicle **200** is provided from the operation terminal **100** to the terminal user UR, the terminal user UR can obtain much information on the state of controlling the own vehicle **200**. Thereby, the terminal user UR has benefits.

(199) Accordingly, the vehicle ECU **290** is configured to wirelessly transmit a moving vibration request signal **S41** via the vehicle signal receiving-and-transmitting device **280** when the own vehicle **200** is autonomously moved by the remote moving control, i.e., the own vehicle **200** is autonomously moved forward or rearward. The moving vibration request signal **S41** is a signal which requests to activate the vibration device **130** to vibrate the operation terminal **100** with a predetermined vibration pattern or a moving vibration pattern. In other words, the vehicle ECU **290** is configured to wirelessly transmit the moving vibration request signal **S41** via the vehicle signal receiving-and-transmitting device **280** when the state of controlling the own vehicle **200** corresponds to a state of moving the own vehicle **200** while the remote moving control is executed.

(200) In this case, when the terminal ECU **190** receives the moving vibration request signal **S41** via the terminal signal receiving-and-transmitting device **180**, the terminal ECU **190** executes a moving vibration process, i.e., the vibration process to activate the vibration device **130** to generate vibration of the moving vibration pattern. In other words, when the state of controlling the own vehicle **200** corresponds to the state of moving the own vehicle **200** while the remote moving control is executed, the terminal ECU **190** executes the moving vibration process.

(201) Thereby, the terminal user UR can realize that the own vehicle **200** moves from the vibration of the operation terminal **100** and with their eyes.

(202) In this embodiment, the vibration pattern is defined by a vibration duration (i.e., a duration period of time of continuously generating the vibration), the vibration number of times (i.e., the number of times of generating the vibration by one vibration process), a vibration interval (i.e., a time interval of generating the vibrations in one vibration process, a vibration intensity (i.e., an intensity of the vibration), and a vibration frequency (i.e., a frequency of the vibration). Thus, the vibration pattern can be changed by changing one or more of the vibration duration, the vibration number of times, the vibration interval, the vibration intensity, and the vibration frequency.

(203) Further, in this embodiment, the vibration pattern of the operation terminal **100** when the own vehicle **200** is moved forward by the remote moving control is the same as the vibration pattern of the operation terminal **100** when the own vehicle **200** is moved rearward by the remote moving control. In this regard, the vibration pattern of the operation terminal **100** when the own vehicle **200** is moved forward by the remote moving control and the vibration pattern of the operation terminal **100** when the own vehicle **200** is moved rearward by the remote moving control may be different.

(204) Furthermore, the vehicle ECU **290** may be configured to wirelessly transmit a moving informing sound request signal **S51** via the vehicle signal receiving-and-transmitting device **280** in addition to or in place of wirelessly transmitting the moving vibration request signal **S41** when the own vehicle **200** is autonomously moved (i.e., the own vehicle **200** is autonomously moved forward or rearward) by the remote moving control. The moving informing sound request signal **S51** is a signal which requests to output informing sounds from the sound device **140** with a predetermined informing sound output pattern or a moving informing sound output pattern. In this case, when the terminal ECU **190** receives the moving informing sound request signal **S51** via the terminal signal receiving-and-transmitting device **180**, the terminal ECU **190** executes a moving informing process, i.e., an informing process to output the informing sounds from the sound device **140** with the moving informing sound output pattern.

(205) Further, the vehicle ECU **290** is configured not to wirelessly transmit the vibration request

signal **S40**, i.e., the signal which requests the vibration by the vibration device **130** when the own vehicle **200** is stopped by the remote moving control. In other words, the vehicle ECU **290** is configured not to wirelessly the vibration request signal **S40** when the state of controlling the own vehicle **200** corresponds to the state of stopping the own vehicle **200** while the remote moving control is executed.

(206) On the other hand, the terminal ECU **190** does not activate the vibration device **130** when the terminal ECU **190** does not receive the vibration request signal **S40**. That is, the terminal ECU **190** does not execute the vibration process. In other words, the terminal ECU **190** does not execute the vibration process when the state of controlling the own vehicle **200** corresponds to the state of stopping the own vehicle **200** while the remote moving control is executed.

(207) Further, the vehicle ECU **290** is configured to wirelessly transmit a shift change vibration request signal **S42** via the vehicle signal receiving-and-transmitting device **280** when the vehicle ECU **290** changes the shift state of the transmission apparatus **224** from the forward moving driving state **SD** to the rearward moving driving state **SR** to K-turn the own vehicle **200**. The shift change vibration request signal **S42** is a signal which requests to vibrate the operation terminal **100** with a predetermined vibration pattern or a shift change vibration pattern by activating the vibration device **130**. In other words, the vehicle ECU **290** is configured to wirelessly transmit the shift change vibration request signal **S42** via the vehicle signal receiving-and-transmitting device **280** when the state of controlling the own vehicle **200** corresponds to a state of changing the shift state of the transmission apparatus **224** while the remote moving control is executed.

(208) It should be noted that the shift change vibration pattern is different from the moving vibration pattern. Further, the vehicle ECU **290** may be configured to wirelessly transmit the shift change vibration request signal **S42** via the vehicle signal receiving-and-transmitting device **280** when the vehicle ECU **290** changes the shift state of the transmission apparatus **224** from the rearward moving driving state **SR** to the forward moving driving state **SD** to K-turn the own vehicle **200**.

(209) Further, the vehicle ECU **290** may be configured to wirelessly transmit the shift change vibration request signal **S42** via the vehicle signal receiving-and-transmitting device **280** when the vehicle ECU **290** changes the shift state of the transmission apparatus **224** from the forward moving driving state **SD** to the rearward moving driving state **SR** or from the rearward moving driving state **SR** to the forward moving driving state **SD** to achieve an object other than an object of K-turning the own vehicle **200**.

(210) In this case, when the terminal ECU **190** receives the shift change vibration request signal **S42** via the terminal signal receiving-and-transmitting device **180**, the terminal ECU **190** executes a shift change vibration process, i.e., the vibration process to activate the vibration device **130** to generate the vibration of the shift change vibration pattern. In other words, the terminal ECU **190** is configured to execute the shift change vibration process when the state of controlling the own vehicle **200** corresponds to the state of changing the shift state of the transmission apparatus **224** while the remote moving control is executed.

(211) It is difficult for the terminal user **UR** to realize that the shift state of the transmission apparatus **224** is changed to K-turn the own vehicle **200** and etc. with their eyes. According to the embodiment, the terminal user **UR** can realize that the shift state of the transmission apparatus **224** is changed by the vibration of the operation terminal **100**.

(212) It should be noted that the vehicle ECU **290** may be configured to wirelessly transmit a shift change informing sound request signal **S52** via the vehicle signal receiving-and-transmitting device **280** in addition to or in place of wirelessly transmitting the shift change vibration request signal **S42** when the vehicle ECU **290** changes the shift state of the transmission apparatus **224**. The shift change informing sound request signal **S52** is a signal which requests to output the informing sounds from the sound device **140** with a predetermined informing sound output pattern or a shift change informing sound output pattern. In this case, the terminal ECU **190** is configured to execute

a shift change informing process, i.e., an informing process to output the informing sounds from the sound device **140** with the shift change informing sound output pattern when the terminal ECU **190** receives the shift change informing sound request signal **S52** via the terminal signal receiving-and-transmitting device **180**. In this regard, the shift change informing sound output pattern is different from the moving informing sound output pattern.

(213) Further, the informing sound output pattern is defined by an informing duration (i.e., a duration period of time of continuously outputting the informing sound), the informing number of times (i.e., the number of times of outputting the informing sound by one informing process), an informing interval (i.e., a time interval of outputting the informing sounds in one informing process), an informing sound intensity (i.e., an intensity of the informing sound), and an informing sound frequency (i.e., a frequency of the informing sound). Thus, the informing sound output pattern can be changed by changing one or more of the informing duration, the informing number of times, the informing interval, the informing sound intensity, and the informing sound frequency.

(214) According to the remote moving system described above, the vibration is generated in the operation terminal **100**, for example, as shown in FIG. **27** while the remote moving control is executed. In an example shown in FIG. **27**, while the remote moving control is executed, at a time **t270**, the own vehicle **200** starts to be moved forward. Then, at a time **t271**, the own vehicle **200** is stopped to K-turn the own vehicle **200**. Then, at a time **t272**, the shift state of the transmission apparatus **224** is changed from the forward moving driving state **SD** to the rearward moving driving state **SR**. Then, at a time **t273**, the own vehicle **200** starts to be moved rearward.

(215) In the example shown in FIG. **27**, when the own vehicle **200** starts to be moved forward at the time **270**, the moving vibration process is started. Thus, the vibration of the moving vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**.

(216) Then, when the own vehicle **200** is stopped at the time **t271**, the vibration process or the moving vibration process is stopped.

(217) Then, when the shift state of the transmission apparatus **224** is changed from the forward moving driving state **SD** to the rearward moving driving state **SR** at the time **t272**, the shift change vibration process is executed, and the vibration of the shift change vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**.

(218) Then, when the own vehicle **200** starts to be moved rearward at the time **t273**, the moving vibration process is started, and the vibration of the moving vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**.

(219) It should be noted that the terminal user **UR** can have benefits if the terminal user **UR** is informed that the own vehicle **200** is braked by the remote moving control.

(220) Accordingly, the vehicle ECU **290** may be configured to wirelessly transmit a braking vibration request signal **S43** via the vehicle signal receiving-and-transmitting device **280** when the vehicle ECU **290** brakes the own vehicle **200** by the remote moving control. The braking vibration request signal **S43** is a signal which requests to vibrate the operation terminal **100** with a predetermined vibration pattern or a braking vibration pattern by activating the vibration device **130**. In other words, the vehicle ECU **290** may be configured to wirelessly transmit the braking vibration request signal **S43** via the vehicle signal receiving-and-transmitting device **280** when the state of controlling the own vehicle **200** corresponds to a state of braking the own vehicle **200** while the remote moving control is executed.

(221) In this case, when the terminal ECU **190** receives the braking vibration request signal **S43** via the terminal signal receiving-and-transmitting device **180**, the terminal ECU **190** executes a braking vibration process, i.e., the vibration process to activate the vibration device **130** to generate the vibration of the braking vibration pattern. In other words, when the state of controlling the own vehicle **200** corresponds to the state of braking the own vehicle **200** while the remote moving control is executed, the terminal ECU **190** executes the braking vibration process. The braking

vibration pattern is different from the moving vibration pattern and the shift change vibration pattern.

(222) Thereby, the terminal user UR can realize that the own vehicle **200** is braked from the vibration of the operation terminal **100** when the terminal user UR cannot easily realize that the own vehicle **200** is braked with their eyes.

(223) It should be noted that the vehicle ECU **290** may be configured to wirelessly transmit a braking informing sound request signal **S53** via the vehicle signal receiving-and-transmitting device **280** in addition to or in place of wirelessly transmitting the braking vibration request signal **S43** when the own vehicle **200** is braked by the remote moving control. The braking informing sound request signal **S53** is a signal which requests to output the informing sounds from the sound device **140** with a predetermined informing sound output pattern or a braking informing sound output pattern. In this case, when the terminal ECU **190** receives the braking informing sound request signal **S53** via the terminal signal receiving-and-transmitting device **180**, the terminal ECU **190** executes a braking informing process, i.e., the informing process to output the informing sounds from the sound device **140** with the braking informing sound output pattern. The braking informing sound output pattern is different from the moving informing sound output pattern and the shift change informing sound output pattern.

(224) Further, the terminal user UR can have benefits if the terminal user UR is informed that an obstacle such as a person which interrupts or may interrupt the moving of the own vehicle **200** is detected, based on the surrounding detection information IS while the own vehicle **200** is moved by the remote moving control.

(225) Accordingly, the vehicle ECU **290** may be configured to wirelessly transmit an obstacle vibration request signal **S44** via the vehicle signal receiving-and-transmitting device **280** when the vehicle ECU **290** detects the obstacle while moving the own vehicle **200** by the remote moving control. The obstacle vibration request signal **S44** is a signal which requests to vibrate the operation terminal **100** with a predetermined vibration pattern or an obstacle vibration pattern by activating the vibration device **130**.

(226) In this case, when the terminal ECU **190** receives the obstacle vibration request signal **S44** via the terminal signal receiving-and-transmitting device **180**, the terminal ECU **190** executes an obstacle vibration process, i.e., the vibration process to activate the vibration device **130** to generate the vibration of the obstacle vibration pattern.

(227) The obstacle vibration pattern is different from the moving vibration pattern, the shift change vibration pattern, and the braking vibration pattern. Further, the obstacle vibration process is executed without executing the braking vibration process even when the own vehicle **200** starts to be braked in response to detecting the obstacle.

(228) Thereby, the terminal user UR can realize that there is the obstacle from the vibration of the operation terminal **100** and with their eyes.

(229) It should be noted that the vehicle ECU **290** may be configured to wirelessly transmit an obstacle informing sound request signal **S54** via the vehicle signal receiving-and-transmitting device **280** in addition to or in place of wirelessly transmitting the obstacle vibration request signal **S44** when the vehicle ECU **290** detects the obstacle while moving the own vehicle **200** by the remote moving control. The obstacle informing sound request signal **S54** is a signal which requests to output the informing sounds from the sound device **140** with a predetermined informing sound output pattern or an obstacle informing sound output pattern. In this case, when the terminal ECU **190** receives the obstacle informing sound request signal **S54** via the terminal signal receiving-and-transmitting device **180**, the terminal ECU **190** executes an obstacle informing process, i.e., the informing process to output the informing sounds from the sound device **140** with the obstacle informing sound output pattern. The obstacle informing sound output pattern is different from the moving informing sound output pattern, the shift change informing sound output pattern, and the braking informing sound output pattern.

(230) According to the remote moving system described above, the vibration is generated in the operation terminal **100**, for example, as shown in FIG. **28** while the remote moving control is executed. In an example shown in FIG. **28**, while the remote moving control is executed, at a time **t280**, the own vehicle **200** starts to be moved forward. Then, at a time **t281**, the own vehicle **200** starts to be braked to K-turn the own vehicle **200**. Then, at a time **t282**, the own vehicle **200** is stopped. Then, at a time **t283**, the shift state of the transmission apparatus **224** is changed from the forward moving driving state SD to the rearward moving driving state SR. Then, at a time **t284**, the own vehicle **200** starts to be moved rearward. Then, at a time **t285**, the own vehicle **200** starts to be braked in response to detecting the obstacle. Then, at a time **286**, the own vehicle **200** is stopped. Then, at a time **287**, the own vehicle **200** starts to be moved rearward after the obstacle disappeared.

(231) In the example shown in FIG. **28**, when the own vehicle **200** starts to be moved forward at the time **280**, the moving vibration process is started. Thus, the vibration of the moving vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**.

(232) Then, when the own vehicle **200** starts to be braked at the time **t281**, the moving vibration process is stopped, the braking vibration process is started, and the vibration of the braking vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**.

(233) Then, when the own vehicle **200** is stopped at the time **t282**, the braking vibration process is stopped.

(234) Then, when the shift state of the transmission apparatus **224** is changed from the forward moving driving state SD to the rearward moving driving state SR at the time **t283**, the shift change vibration process is executed, and the vibration of the shift change vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**.

(235) Then, when the own vehicle **200** starts to be moved rearward at the time **t284**, the moving vibration process is started, and the vibration of the moving vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**.

(236) Then, when the own vehicle **200** starts to be braked in response to detecting the obstacle at the time **285**, the moving vibration process is stopped, the obstacle vibration process is started, and the vibration of the obstacle vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**. Then, at the time **t286**, the own vehicle **200** is stopped.

(237) Then, when the obstacle disappeared, and the own vehicle **200** starts to be moved at the time **t287**, the obstacle vibration process is stopped, the moving vibration process is started, and the vibration of the moving vibration pattern is generated in the operation terminal **100** by the vibration device **130** of the operation terminal **100**.

(238) As described above, the remote moving system **10** executes the moving vibration process while the own vehicle **200** is moved forward or rearward by the remote moving control and executes the shift change vibration process when the own vehicle **200** is K-turned. In addition to these vibration processes, the remote moving system **10** may execute the obstacle vibration process when the obstacle is detected. In this regard, if there is information to be informed to the terminal user UR by the vibration of the operation terminal **100** other than the information on (i) the moving state of the own vehicle **200** (i.e., whether the own vehicle **200** is moved forward or rearward), (ii) the K-turn of the own vehicle **200**, and (iii) a detection of the obstacle, the remote moving system **10** may be configured to inform the terminal user UR of the other information by vibrating the operation terminal **100** by the vibration device **130** with the vibration pattern different from the vibration patterns used to inform the terminal user UR of the moving state of the own vehicle **200**, the K-turn of the own vehicle **200**, and the detection of the obstacle.

(239) The remote moving system **10** has been described. According to the remote moving system

10, the operation terminal **100** is vibrated with the different vibration patterns when the own vehicle **200** is moved by the remote moving control, when the own vehicle **200** is braked, when the shift state of the transmission apparatus **224** is changed, and when the obstacle is detected. Thus, the terminal user UR can realize the state of controlling the own vehicle **200** and the obstacle from the vibration of the operation terminal **100** while watching the own vehicle **200** without watching the terminal display **120**. In other words, the vibration patterns used are different, depending on the state of controlling the own vehicle **200** and a situation around the own vehicle **200**. Thus, the terminal user UR can realize the state of controlling the own vehicle **200** and the obstacle from the vibration of the operation terminal **100** while watching the own vehicle **200** without watching the terminal display **120**.

(240) <Specific Operations of Remote Moving System>

(241) Next, specific operations of the remote moving system will be described. The terminal CPU **191** of the terminal ECU **190** of the terminal control unit **110** is configured or programmed to execute a routine shown in FIG. **29** with a predetermined calculation cycle. Thus, at a predetermined timing, the terminal CPU **191** starts a process from a step **2900** of the routine shown in FIG. **29** and proceeds with the process to a step **2905** to determine whether a value of an initial screen image display flag X**10** is “0”. The initial screen image display flag X**10** is a flag which represents that the initial screen image G**10** is displayed on the terminal display **120**. The value of the initial screen image display flag X**10** is set to “1” when the initial screen image G**10** is displayed on the terminal display **120**. When the value of the initial screen image display flag X**10** is “0”, the initial screen image G**10** is not displayed on the terminal display **120**.

(242) When the terminal CPU **191** determines “Yes” at the step **2905**, the terminal CPU **191** proceeds with the process to a step **2910** to determine whether the application start operation is carried out. That is, the terminal CPU **191** determines whether the predetermined touch operation to start the remote moving application software is applied to the terminal display **120**. When the terminal CPU **191** determines “Yes” at the step **2910**, the terminal CPU **191** proceeds with the process to a step **2915** to start the remote moving application software and display the initial screen image G**10**. Next, the terminal CPU **191** proceeds with the process to a step **2920** to wirelessly transmit the application start signal S**10**. Next, the terminal CPU **191** proceeds with the process to a step **2925** to set the value of the initial screen image display flag X**10** to “1”. Next, the terminal CPU **191** proceeds with the process to a step **2995** to terminate this routine once.

(243) On the other hand, when the terminal CPU **191** determines “No” at the step **2910**, the terminal CPU **191** proceeds with the process directly to the step **2995** to terminate this routine once.

(244) Further, when the terminal CPU **191** determines “No” at the step **2905**, the terminal CPU **191** proceeds with the process to a step **2930** to determine whether an application termination operation is carried out. That is, the terminal CPU **191** determines whether the tap operation is applied to any one of the application termination image portions P**15** to P**65**. When the terminal CPU **191** determines “Yes” at the step **2930**, the terminal CPU **191** proceeds with the process to a step **2935** to terminate the remote moving application software. Next, the terminal CPU **191** proceeds with the process to a step **2940** to wirelessly transmit the control termination command signal S**30**. Next, the terminal CPU **191** proceeds with the process to a step **2945** to set the value of the initial screen image display flag X**10** to “0” and set values of a parking-and-pulling-out selection image display flag X**11**, a parking space selection image display flag X**12**, a pulling-out direction selection image display flag X**13**, a parking operation image display flag X**14**, and a pulling-out operation image display flag X**15** to “0”, respectively. Next, the terminal CPU **191** proceeds with the process to the step **2995** to terminate this routine once.

(245) On the other hand, when the terminal CPU **191** determines “No” at the step **2930**, the terminal CPU **191** proceeds with the process directly to the step **2995** to terminate this routine once.

(246) Further, the terminal CPU **191** is configured or programmed to execute a routine shown in FIG. **30** with the predetermined calculation cycle. Thus, at a predetermined timing, the terminal CPU **191** starts a process from a step **3000** of the routine shown in FIG. **30** and proceeds with the process to a step **3005** to determine whether the value of the initial screen image display flag **X10** is “1”. When the terminal CPU **191** determines “Yes” at the step **3005**, the terminal CPU **191** proceeds with the process to a step **3010** to determine whether the value of the parking-and-pulling-out selection image display flag **X11** is “0”. The parking-and-pulling-out selection image display flag **X11** is a flag which represents that the parking-and-pulling-out selection image **G20** is displayed on the terminal display **120**. The value of the parking-and-pulling-out selection image display flag **X11** is set to “1” when the parking-and-pulling-out selection image **G20** is displayed on the terminal display **120**. When the value of the parking-and-pulling-out selection image display flag **X11** is “0”, the parking-and-pulling-out selection image **G20** is not displayed on the terminal display **120**.

(247) When the terminal CPU **191** determines “Yes” at the step **3010**, the terminal CPU **191** proceeds with the process to a step **3015** to determine whether a remote parking-and-pulling-out start operation is carried out. That is, the terminal CPU **191** determines whether the touch operation is applied to the remote parking-and-pulling-out start image portion **P11**. When the terminal CPU **191** determines “Yes” at the step **3015**, the terminal CPU **191** proceeds with the process to a step **3020** to display the parking-and-pulling-out selection image **G20** on the terminal display **120**. Next, the terminal CPU **191** proceeds with the process to a step **3025** to set the value of the parking-and-pulling-out selection image display flag **X11** to “1”. Next, the terminal CPU **191** proceeds with the process to a step **3095** to terminate this routine once.

(248) On the other hand, when the terminal CPU **191** determines “No” at the step **3015**, the terminal CPU **191** proceeds with the process directly to the step **3095** to terminate this routine once.

(249) Further, when the terminal CPU **191** determines “No” at the step **3005** or **3010**, the terminal CPU **191** proceeds with the process directly to the step **3095** to terminate this routine once.

(250) Furthermore, the terminal CPU **191** is configured or programmed to execute a routine shown in FIG. **31** with the predetermined calculation cycle. Thus, at a predetermined timing, the terminal CPU **191** starts a process from a step **3100** of the routine shown in FIG. **31** and proceeds with the process to a step **3105** to determine whether the value of the initial screen image display flag **X11** is “1”. When the terminal CPU **191** determines “Yes” at the step **3105**, the terminal CPU **191** proceeds with the process to a step **3110** to determine whether the value of the parking space selection image display flag **X12** is “0”. The parking space selection image display flag **X12** is a flag which represents that the parking space selection image **G30** is displayed on the terminal display **120**. The value of the parking space selection image display flag **X12** is set to “1” when the parking space selection image **G30** is displayed on the terminal display **120**. When the value of the parking space selection image display flag **X12** is “0”, the parking space selection image **G30** is not displayed on the terminal display **120**.

(251) When the terminal CPU **191** determines “Yes” at the step **3110**, the terminal CPU **191** proceeds with the process to a step **3115** to determine whether a parking selection operation is carried out. That is, the terminal CPU **191** determines whether the touch operation is applied to the parking selection image portion **P21**. When the terminal CPU **191** determines “Yes” at the step **3115**, the terminal CPU **191** proceeds with the process to a step **3120** to display the parking space selection image **G30** on the terminal display **120**. Next, the terminal CPU **191** proceeds with the process to a step **3125** to set the value of the parking space selection image display flag **X12** to “1”. Next, the terminal CPU **191** proceeds with the process to a step **3130**.

(252) On the other hand, when the terminal CPU **191** determines “No” at the step **3115**, the terminal CPU **191** proceeds with the process directly to the step **3130**.

(253) When the terminal CPU **191** proceeds with the process to the step **3130**, the terminal CPU

191 determines whether the value of the pulling-out direction selection image display flag **X13** is “0”. The pulling-out direction selection image display flag **X13** is a flag which represents that the pulling-out direction selection image **G50** is displayed on the terminal display **120**. The value of the pulling-out direction selection image display flag **X13** is set to “1” when the pulling-out direction selection image **G50** is displayed on the terminal display **120**. When the value of the pulling-out direction selection image display flag **X13** is “0”, the pulling-out direction selection image **G50** is not displayed on the terminal display **120**.

(254) When the terminal CPU **191** determines “Yes” at the step **3130**, the terminal CPU **191** proceeds with the process to a step **3135** to determine whether a pulling-out selection operation is carried out. That is, the terminal CPU **191** determines whether the touch operation is applied to the pulling-out selection image portion **P22**. When the terminal CPU **191** determines “Yes” at the step **3135**, the terminal CPU **191** proceeds with the process to a step **3140** to display the pulling-out direction selection image **G50** on the terminal display **120**. Next, the terminal CPU **191** proceeds with the process to a step **3145** to set the value of the pulling-out direction selection image display flag **X13** to “1”. Next, the terminal CPU **191** proceeds with the process to a step **3195** to terminate this routine once.

(255) On the other hand, when the terminal CPU **191** determines “No” at the step **3135**, the terminal CPU **191** proceeds with the process directly to the step **3195** to terminate this routine once.

(256) Further, when the terminal CPU **191** determines “No” at the step **3105** or **3110** or **3130**, the terminal CPU **191** proceeds with the process directly to the step **3195** to terminate this routine once.

(257) Furthermore, the terminal CPU **191** is configured or programmed to execute a routine shown in FIG. **32** with the predetermined calculation cycle. Thus, at a predetermined timing, the terminal CPU **191** starts a process from a step **3200** of the routine shown in FIG. **32** and proceeds with the process to a step **3205** to determine whether the value of the parking space selection image display flag **X12** is “1”. When the terminal CPU **191** determines “Yes” at the step **3205**, the terminal CPU **191** proceeds with the process to a step **3210** to determine whether the value of the parking operation image display flag **X14** is “0”. The parking operation image display flag **X14** is a flag which represents that the parking operation image **G40** is displayed on the terminal display **120**. The value of the parking operation image display flag **X14** is set to “1” when the parking operation image **G40** is displayed on the terminal display **120**. When the value of the parking operation image display flag **X14** is “0”, the parking operation image **G40** is not displayed on the terminal display **120**.

(258) When the terminal CPU **191** determines “Yes” at the step **3210**, the terminal CPU **191** proceeds with the process to a step **3215** to determine whether a parking space selection operation is applied to the terminal display **120**. That is, the terminal CPU **191** determines whether the touch operation is applied to the available parking space image portion **P31**. When the terminal CPU **191** determines “Yes” at the step **3215**, the terminal CPU **191** proceeds with the process to a step **3220** to determine whether a set operation is applied to the terminal display **120**. That is, the terminal CPU **191** determines whether the touch operation is applied to the selection set image portion **P34**.

(259) When the terminal CPU **191** determines “Yes” at the step **3220**, the terminal CPU **191** proceeds with the process to a step **3225** to display the parking operation image **G40** on the terminal display **120**. Next, the terminal CPU **191** proceeds with the process to a step **3230** to wirelessly transmit the designated parking space signal **S12**. At this time, the terminal CPU **191** displays the vehicle image **G42** on the terminal display **120** at a position corresponding to the position of the own vehicle **200** represented by the vehicle position signal **S13**, displays the parking remaining distance image **G43** displaying the parking remaining distance **Din** represented by the parking remaining distance signal **S15** on the terminal display **120**, and displays the moving direction image **G47** displaying the moving direction of the own vehicle **200** represented by the

moving direction signal S17 on the terminal display 120.

(260) Next, the terminal CPU 191 proceeds with the process to a step 3235 to set the value of the parking operation image display flag X14 to "1". Next, the terminal CPU 191 proceeds with the process to a step 3240.

(261) On the other hand, when the terminal CPU 191 determines "No" at the step 3220, the terminal CPU 191 proceeds with the process directly to the step 3240.

(262) Further, when the terminal CPU 191 determines "No" at the step 3215, the terminal CPU 191 proceeds with the process directly to the step 3240.

(263) When the terminal CPU 191 proceeds with the process to the step 3240, the terminal CPU 191 determines whether a parking-and-pulling-out reselection operation is applied to the terminal display 120. That is, the terminal CPU 191 determines whether the touch operation is applied to the parking-and-pulling-out selection image portion P36. When the terminal CPU 191 determines "Yes" at the step 3240, the terminal CPU 191 proceeds with the process to a step 3245 to display the parking-and-pulling-out selection image G20 on the terminal display 120. Next, the terminal CPU 191 proceeds with the process to a step 3250 to set the values of the parking-and-pulling-out selection image display flag X11, the parking space selection image display flag X12, and the parking operation image display flag X14 to "0", respectively. Next, the terminal CPU 191 proceeds with the process to a step 3295 to terminate this routine once.

(264) On the other hand, when the terminal CPU 191 determines "No" at the step 3240, the terminal CPU 191 proceeds with the process directly to the step 3295 to terminate this routine once.

(265) Further, when the terminal CPU 191 determines "No" at the step 3205 or 3210, the terminal CPU 191 proceeds with the process directly to the step 3295 to terminate this routine once.

(266) Furthermore, the terminal CPU 191 is configured or programmed to execute a routine shown in FIG. 33 with the predetermined calculation cycle. Thus, at a predetermined timing, the terminal CPU 191 starts a process from a step 3300 of the routine shown in FIG. 33 and proceeds with the process to a step 3305 to determine whether the value of the pulling-out direction selection image display flag X13 is "1". When the terminal CPU 191 determines "Yes" at the step 3305, the terminal CPU 191 proceeds with the process to a step 3310 to determine whether the value of the pulling-out operation image display flag X15 is "0". The pulling-out operation image display flag X15 is a flag which represents that the pulling-out operation image G60 is displayed on the terminal display 120. The value of the pulling-out operation image display flag X15 is set to "1" when the pulling-out operation image G60 is displayed on the terminal display 120. When the value of the pulling-out operation image display flag X15 is "0", the pulling-out operation image G60 is not displayed on the terminal display 120.

(267) When the terminal CPU 191 determines "Yes" at the step 3310, the terminal CPU 191 proceeds with the process to a step 3315 to determine whether a pulling-out direction selection operation is applied to the terminal display 120. That is, the terminal CPU 191 determines whether the touch operation is applied to the selectable pulling-out direction image portion P53. When the terminal CPU 191 determines "Yes" at the step 3315, the terminal CPU 191 proceeds with the process to a step 3320 to determine whether the set operation is applied to the terminal display 120. That is, the terminal CPU 191 determines whether the touch operation is applied to the selection set image portion P54.

(268) When the terminal CPU 191 determines "Yes" at the step 3320, the terminal CPU 191 proceeds with the process to a step 3325 to display the pulling-out operation image G60 on the terminal display 120. Next, the terminal CPU 191 proceeds with the process to a step 3330 to wirelessly transmit the designated pulling-out direction signal S22. At this time, the terminal CPU 191 displays the vehicle image G62 on the terminal display 120 at a position corresponding to the position of the own vehicle 200 represented by the vehicle position signal S23, and displays the pulling-out remaining distance image G63 displaying the parking-out remaining distance D_{out}

represented by the pulling-out remaining distance signal S25 on the terminal display 120.

(269) Next, the terminal CPU 191 proceeds with the process to a step 3335 to set the value of the pulling-out operation image display flag X15 to “1”. Next, the terminal CPU 191 proceeds with the process to a step 3340.

(270) On the other hand, when the terminal CPU 191 determines “No” at the step 3320, the terminal CPU 191 proceeds with the process directly to the step 3340.

(271) Further, when the terminal CPU 191 determines “No” at the step 3315, the terminal CPU 191 proceeds with the process directly to the step 3340.

(272) When the terminal CPU 191 proceeds with the process to the step 3340, the terminal CPU 191 determines whether the parking-and-pulling-out reselection operation is applied to the terminal display 120. That is, the terminal CPU 191 determines whether the touch operation is applied to the parking-and-pulling-out reselection image portion P56. When the terminal CPU 191 determines “Yes” at the step 3340, the terminal CPU 191 proceeds with the process to a step 3345 to display the parking-and-pulling-out selection image G20 on the terminal display 120. Next, the terminal CPU 191 proceeds with the process to a step 3350 to set the values of the parking-and-pulling-out selection image display flag X11, the pulling-out direction selection image display flag X13, and the pulling-out operation image display flag X15 to “0”, respectively. Next, the terminal CPU 191 proceeds with the process to a step 3395 to terminate this routine once.

(273) On the other hand, when the terminal CPU 191 determines “No” at the step 3340, the terminal CPU 191 proceeds with the process directly to the step 3395 to terminate this routine once.

(274) Further, when the terminal CPU 191 determines “No” at the step 3305 or 3310, the terminal CPU 191 proceeds with the process directly to the step 3395 to terminate this routine once.

(275) Furthermore, the terminal CPU 191 is configured or programmed to execute a routine shown in FIG. 34 with the predetermined calculation cycle. Thus, at a predetermined timing, the terminal CPU 191 starts a process from a step 3400 of the routine shown in FIG. 34 and proceeds with the process to a step 3405 to determine whether the value of the parking operation image display flag X14 is “1”. When the terminal CPU 191 determines “Yes” at the step 3405, the terminal CPU 191 proceeds with the process to a step 3410 to determine whether a moving operation is carried out. That is, the terminal CPU 191 determines whether the touch operation is applied to the parking moving operation image portion P44. When the terminal CPU 191 determines “Yes” at the step 3410, the terminal CPU 191 proceeds with the process to a step 3415 to wirelessly transmit the continuous touch operation signal S16. Next, the terminal CPU 191 proceeds with the process to a step 3420.

(276) On the other hand, when the terminal CPU 191 determines “No” at the step 3410, the terminal CPU 191 proceeds with the process directly to the step 3420.

(277) When the terminal CPU 191 proceeds with the process to the step 3420, the terminal CPU 191 determines whether parking the own vehicle 200 in the designated parking space 31D is completed. That is, the terminal CPU 191 determines whether to have received the parking complete signal S18. When the terminal CPU 191 determines “Yes” at the step 3420, the terminal CPU 191 proceeds with the process to a step 3425 to display the parking complete image G49 on the terminal display 120. Next, the terminal CPU 191 proceeds with the process to a step 3430.

(278) On the other hand, when the terminal CPU 191 determines “No” at the step 3420, the terminal CPU 191 proceeds with the process directly to the step 3430.

(279) When the terminal CPU 191 proceeds with the process to the step 3440, the terminal CPU 191 determines whether the parking-and-pulling-out reselection operation is applied to the terminal display 120. That is, the terminal CPU 191 determines whether the touch operation is applied to the parking-and-pulling-out reselection image portion P46. When the terminal CPU 191 determines “Yes” at the step 3430, the terminal CPU 191 proceeds with the process to a step 3435 to display the parking-and-pulling-out selection image G20 on the terminal display 120. Next, the terminal

CPU **191** proceeds with the process to a step **3440** to set the values of the parking space selection image display flag **X12** and the parking operation image display flag **X14** to "0", respectively. Next, the terminal CPU **191** proceeds with the process to a step **3495** to terminate this routine once. (280) On the other hand, when the terminal CPU **191** determines "No" at the step **3430**, the terminal CPU **191** proceeds with the process directly to the step **3495** to terminate this routine once.

(281) Further, when the terminal CPU **191** determines "No" at the step **3405**, the terminal CPU **191** proceeds with the process directly to the step **3495** to terminate this routine once.

(282) Furthermore, the terminal CPU **191** is configured or programmed to execute a routine shown in FIG. **35** with the predetermined calculation cycle. Thus, at a predetermined timing, the terminal CPU **191** starts a process from a step **3500** of the routine shown in FIG. **35** and proceeds with the process to a step **3505** to determine whether the value of the pulling-out operation image display flag **X15** is "1". When the terminal CPU **191** determines "Yes" at the step **3505**, the terminal CPU **191** proceeds with the process to a step **3510** to determine whether the moving operation is carried out. That is, the terminal CPU **191** determines whether the touch operation is applied to the pulling-out moving operation image portion **P64**. When the terminal CPU **191** determines "Yes" at the step **3510**, the terminal CPU **191** proceeds with the process to a step **3515** to wirelessly transmit the continuous touch operation signal **S26**. Next, the terminal CPU **191** proceeds with the process to a step **3520**.

(283) On the other hand, when the terminal CPU **191** determines "No" at the step **3510**, the terminal CPU **191** proceeds with the process directly to the step **3520**.

(284) When the terminal CPU **191** proceeds with the process to the step **3520**, the terminal CPU **191** determines whether pulling out the own vehicle **200** from the current parking space **31N** is completed. That is, the terminal CPU **191** determines whether to have received the pulling-out complete signal **S28**. When the terminal CPU **191** determines "Yes" at the step **3520**, the terminal CPU **191** proceeds with the process to a step **3525** to display the pulling-out complete image **G69** on the terminal display **120**. Next, the terminal CPU **191** proceeds with the process to a step **3530**.

(285) On the other hand, when the terminal CPU **191** determines "No" at the step **3520**, the terminal CPU **191** proceeds with the process directly to the step **3530**.

(286) When the terminal CPU **191** proceeds with the process to the step **3530**, the terminal CPU **191** determines whether the parking-and-pulling-out reselection operation is applied to the terminal display **120**. That is, the terminal CPU **191** determines whether the touch operation is applied to the parking-and-pulling-out reselection image portion **P66**. When the terminal CPU **191** determines "Yes" at the step **3530**, the terminal CPU **191** proceeds with the process to a step **3535** to display the parking-and-pulling-out selection image **G20** on the terminal display **120**. Next, the terminal CPU **191** proceeds with the process to a step **3540** to set the values of the pulling-out direction selection image display flag **X13** and the pulling-out operation image display flag **X15** to "0", respectively. Next, the terminal CPU **191** proceeds with the process to a step **3595** to terminate this routine once.

(287) On the other hand, when the terminal CPU **191** determines "No" at the step **3530**, the terminal CPU **191** proceeds with the process directly to the step **3595** to terminate this routine once.

(288) Further, when the terminal CPU **191** determines "No" at the step **3505**, the terminal CPU **191** proceeds with the process directly to the step **3595** to terminate this routine once.

(289) On the other hand, the vehicle CPU **291** of the vehicle ECU **290** of the vehicle control unit **210** is configured or programmed to execute a routine shown in FIG. **36** with the predetermined calculation cycle. Thus, at a predetermined timing, the vehicle CPU **291** starts a process from a step **3600** of the routine shown in FIG. **36** and proceeds with the process to a step **3605** to determine whether a value of a parking route set complete flag **X20** is "0". The parking route set complete flag **X20** is a flag which represents that the target parking route **Rin_tgt** is set. The value of the

parking route set complete flag X20 is set to “1” when the target parking route Rin_tgt is set. When the value of the parking route set complete flag X20 is “0”, the target parking route Rin_tgt is not set.

(290) When the vehicle CPU 291 determines “Yes” at the step 3605, the vehicle CPU 291 proceeds with the process to a step 3610 to determine whether to have received the designated parking space signal S14. When the vehicle CPU 291 determines “Yes” at the step 3610, the vehicle CPU 291 proceeds with the process to a step 3615 to set the target parking route Rin_tgt. Next, the vehicle CPU 291 proceeds with the process to a step 3620 to set the value of the parking route set complete flag X20 to “1”. Next, the vehicle CPU 291 proceeds with the process to a step 3695 to terminate this routine once.

(291) On the other hand, when the vehicle CPU 291 determines “No” at the step 3610, the vehicle CPU 291 proceeds with the process directly to the step 3695 to terminate this routine once.

(292) Further, when the vehicle CPU 291 determines “No” at the step 3605, the vehicle CPU 291 proceeds with the process directly to the step 3695 to terminate this routine once.

(293) Further, the vehicle CPU 291 is configured or programmed to execute a routine shown in FIG. 37 with the predetermined calculation cycle. Thus, at a predetermined timing, the vehicle CPU 291 starts a process from a step 3700 of the routine shown in FIG. 37 and proceeds with the process to a step 3705 to determine whether the value of the parking route set complete flag X20 is “1”. When the vehicle CPU 291 determines “Yes” at the step 3705, the vehicle CPU 291 proceeds with the process to a step 3710 to determine whether to have received the continuous touch operation signal S16.

(294) When the vehicle CPU 291 determines “Yes” at the step 3710, the vehicle CPU 291 proceeds with the process to a step 3715 to control the operations of the vehicle moving apparatus 220 to move the own vehicle 200 along the target parking route Rin_tgt. Next, the vehicle CPU 291 proceeds with the process to a step 3720 to determine whether the own vehicle 200 arrives at the target parking point Pin_tgt. That is, the vehicle CPU 291 determines whether parking the own vehicle 200 in the designated parking space 31D is completed.

(295) When the vehicle CPU 291 determines “Yes” at the step 3720, the vehicle CPU 291 proceeds with the process to a step 3725 to execute the vehicle stop process and the vehicle stop holding process. Next, the vehicle CPU 291 proceeds with the process to a step 3730 to stop the operations of the surrounding information detecting apparatus 260 and the vehicle moving apparatus 220. Next, the vehicle CPU 291 proceeds with the process to a step 3735 to wirelessly transmit the parking complete signal S18. Next, the vehicle CPU 291 proceeds with the process to a step 3740 to set the value of the parking route set complete flag X20 to “0”. Next, the vehicle CPU 291 proceeds with the process to a step 3795 to terminate this routine once.

(296) On the other hand, when the vehicle CPU 291 determines “No” at the step 3720, the vehicle CPU 291 proceeds with the process directly to the step 3795 to terminate this routine once.

(297) Further, when the vehicle CPU 291 determines “No” at the step 3710, the vehicle CPU 291 proceeds with the process to a step 3745 to execute the vehicle stop process. Next, the vehicle CPU 291 proceeds with the process to the step 3795 to terminate this routine once.

(298) Further, when the vehicle CPU 291 determines “No” at the step 3705, the vehicle CPU 291 proceeds with the process directly to the step 3795 to terminate this routine once.

(299) Further, the vehicle CPU 291 is configured or programmed to execute a routine shown in FIG. 38 with the predetermined calculation cycle. Thus, at a predetermined timing, the vehicle CPU 291 starts a process from a step 3800 of the routine shown in FIG. 38 and proceeds with the process to a step 3805 to determine whether a value of a pulling-out route set complete flag X21 is “0”. The pulling-out route set complete flag X21 is a flag which represents that the target pulling-out route Rout_tgt is set. The value of the pulling-out route set complete flag X21 is set to “1” when the target pulling-out route Rout_tgt is set. When the value of the pulling-out route set complete flag X21 is “0”, the target pulling-out route Rout_tgt is not set.

(300) When the vehicle CPU **291** determines “Yes” at the step **3805**, the vehicle CPU **291** proceeds with the process to a step **3810** to determine whether to have received the designated pulling-out direction signal **S22**. When the vehicle CPU **291** determines “Yes” at the step **3810**, the vehicle CPU **291** proceeds with the process to a step **3815** to set the target pulling-out route Rout_tgt. Next, the vehicle CPU **291** proceeds with the process to a step **3820** to set the value of the pulling-out route set complete flag **X21** to “1”. Next, the vehicle CPU **291** proceeds with the process to a step **3895** to terminate this routine once.

(301) On the other hand, when the vehicle CPU **291** determines “No” at the step **3810**, the vehicle CPU **291** proceeds with the process directly to the step **3895** to terminate this routine once.

(302) Further, when the vehicle CPU **291** determines “No” at the step **3805**, the vehicle CPU **291** proceeds with the process directly to the step **3895** to terminate this routine once.

(303) Further, the vehicle CPU **291** is configured or programmed to execute a routine shown in FIG. **39** with the predetermined calculation cycle. Thus, at a predetermined timing, the vehicle CPU **291** starts a process from a step **3900** of the routine shown in FIG. **39** and proceeds with the process to a step **3905** to determine whether the value of the pulling-out route set complete flag **X21** is “1”. When the vehicle CPU **291** determines “Yes” at the step **3905**, the vehicle CPU **291** proceeds with the process to a step **3910** to determine whether to have received the continuous touch operation signal **S26**.

(304) When the vehicle CPU **291** determines “Yes” at the step **3910**, the vehicle CPU **291** proceeds with the process to a step **3915** to control the operations of the vehicle moving apparatus **220** to move the own vehicle **200** along the target pulling-out route Rout_tgt. Next, the vehicle CPU **291** proceeds with the process to a step **3920** to determine whether the own vehicle **200** arrives at the target pulling-out point Pout_tgt. That is, the vehicle CPU **291** determines whether pulling out the own vehicle **200** from the designated parking space **31D** is completed.

(305) When the vehicle CPU **291** determines “Yes” at the step **3920**, the vehicle CPU **291** proceeds with the process to a step **3925** to execute the vehicle stop process and the vehicle stop holding process. Next, the vehicle CPU **291** proceeds with the process to a step **3930** to stop the operations of the surrounding information detecting apparatus **260** and the vehicle moving apparatus **220**. Next, the vehicle CPU **291** proceeds with the process to a step **3935** to wirelessly transmit the pulling-out complete signal **S28**. Next, the vehicle CPU **291** proceeds with the process to a step **3940** to set the value of the pulling-out route set complete flag **X21** to “0”. Next, the vehicle CPU **291** proceeds with the process to a step **3995** to terminate this routine once.

(306) On the other hand, when the vehicle CPU **291** determines “No” at the step **3920**, the vehicle CPU **291** proceeds with the process directly to the step **3995** to terminate this routine once.

(307) Further, when the vehicle CPU **291** determines “No” at the step **3910**, the vehicle CPU **291** proceeds with the process to a step **3945** to execute the vehicle stop process. Next, the vehicle CPU **291** proceeds with the process to the step **3995** to terminate this routine once.

(308) Further, when the vehicle CPU **291** determines “No” at the step **3905**, the vehicle CPU **291** proceeds with the process directly to the step **3995** to terminate this routine once.

(309) Further, the vehicle CPU **291** is configured or programmed to execute a routine shown in FIG. **40** with the predetermined calculation cycle. Thus, at a predetermined timing, the vehicle CPU **291** starts a process from a step **4000** of the routine shown in FIG. **40** and proceeds with the process to a step **4005** to determine whether to have received the control termination command signal **S30**.

(310) When the vehicle CPU **291** determines “Yes” at the step **4005**, the vehicle CPU **291** proceeds with the process to a step **4010** to execute the vehicle stop process and the vehicle stop holding process. Next, the vehicle CPU **291** proceeds with the process to a step **4015** to stop the operations of the surrounding information detecting apparatus **260** and the vehicle moving apparatus **220**. Next, the vehicle CPU **291** proceeds with the process to a step **4020** to set the values of the parking route set complete flag **X20** and the pulling-out route set complete flag **X21** to “0”, respectively.

Next, the vehicle CPU **291** proceeds with the process to a step **4095** to terminate this routine once.

(311) On the other hand, when the vehicle CPU **291** determines “No” at the step **4005**, the vehicle CPU **291** proceeds with the process directly to the step **4095** to terminate this routine once.

(312) Further, the vehicle CPU **291** is configured or programmed to execute a routine shown in FIG. **41** or FIG. **43** with the predetermined calculation cycle. Thus, when the vehicle CPU **291** is configured or programmed to execute the routine shown in FIG. **41**, at a predetermined timing, the vehicle CPU **291** starts a process from a step **4100** of the routine shown in FIG. **41** and proceeds with the process to a step **4105** to determine whether the remote moving control is executed.

(313) When the vehicle CPU **291** determines “Yes” at the step **4105**, the vehicle CPU **291** proceeds with the process to a step **4110** to determine whether the own vehicle **200** is moved forward or rearward. When the vehicle CPU **291** determines “Yes” at the step **4110**, the vehicle CPU **291** proceeds with the process to a step **4115** to wirelessly transmit the moving vibration request signal **S41** via the vehicle signal receiving-and-transmitting device **280**. Next, the vehicle CPU **291** proceeds with the process to a step **4120**.

(314) On the other hand, when the vehicle CPU **291** determines “No” at the step **4110**, the vehicle CPU **291** proceeds with the process directly to the step **4120**.

(315) When the vehicle CPU **291** proceeds with the process to the step **4120**, the vehicle CPU **291** determines whether the shift state of the transmission apparatus **224** is changed from the forward driving state to the rearward driving state or from rearward driving state to the forward driving state. When the vehicle CPU **291** determines “Yes” at the step **4120**, the vehicle CPU **291** proceeds with the process to a step **4125** to wirelessly transmit the shift change vibration request signal **S42** via the vehicle signal receiving-and-transmitting device **280**. Next, the vehicle CPU **291** proceeds with the process to a step **4195** to terminate this routine once.

(316) On the other hand, when the vehicle CPU **291** determines “No” at the step **4120**, the vehicle CPU **291** proceeds with the process directly to the step **4195** to terminate this routine once.

(317) Further, when the vehicle CPU **291** determines “No” at the step **4105**, the vehicle CPU **291** proceeds with the process directly to the step **4195** to terminate this routine once.

(318) On the other hand, the terminal CPU **191** is configured or programmed to execute a routine shown in FIG. **42** with the predetermined calculation cycle when the vehicle CPU **291** is configured or programmed to execute the routine shown in FIG. **41**. Thus, at a predetermined timing, the terminal CPU **191** starts a process from a step **4200** of the routine shown in FIG. **42** and proceeds with the process to a step **4205** to determine whether the remote moving application software is executed.

(319) When the terminal CPU **191** determines “Yes” at the step **4205**, the terminal CPU **191** proceeds with the process to a step **4210** to determine whether to have received the moving vibration request signal **S41**. When the terminal CPU **191** determines “Yes” at the step **4210**, the terminal CPU **191** proceeds with the process to a step **4215** to execute the moving vibration process. Thereby, the operation terminal **100** is vibrated with the moving vibration pattern. Next, the terminal CPU **191** proceeds with the process to a step **4220**.

(320) On the other hand, when the terminal CPU **191** determines “No” at the step **4210**, the terminal CPU **191** proceeds with the process directly to the step **4220**.

(321) When the terminal CPU **191** proceeds with the process to the step **4220**, the terminal CPU **191** determines whether to have received the shift change vibration request signal **S42**. When the terminal CPU **191** determines “Yes” at the step **4220**, the terminal CPU **191** proceeds with the process to a step **4225** to execute the shift change vibration process. Thereby, the operation terminal **100** is vibrated with the shift change vibration pattern. Next, the terminal CPU **191** proceeds with the process to a step **4295** to terminate this routine once.

(322) On the other hand, when the terminal CPU **191** determines “No” at the step **4220**, the terminal CPU **191** proceeds with the process directly to the step **4295** to terminate this routine once.

(323) Further, when the terminal CPU **191** determines “No” at the step **4205**, the terminal CPU **191** proceeds with the process directly to the step **4295** to terminate this routine once.

(324) When the vehicle CPU **291** is configured or programmed to execute the routine shown in FIG. **43**, at a predetermined timing, the vehicle CPU **291** starts a process from a step **4300** of the routine shown in FIG. **43** and proceeds with the process to a step **4305** to determine whether the remote moving control is executed.

(325) When the vehicle CPU **291** determines “Yes” at the step **4305**, the vehicle CPU **291** proceeds with the process to a step **4310** to determine whether the own vehicle **200** is moved forward or rearward. When the vehicle CPU **291** determines “Yes” at the step **4310**, the vehicle CPU **291** proceeds with the process to a step **4315** to determine that the obstacle is detected. When the vehicle CPU **291** determines “Yes” at the step **4315**, the vehicle CPU **291** proceeds with the process to a step **4320** to wirelessly transmit the obstacle vibration request signal **S44** via the vehicle signal receiving-and-transmitting device **280**. Next, the vehicle CPU **291** proceeds with the process to a step **4340**.

(326) On the other hand, when the vehicle CPU **291** determines “No” at the step **4315**, the vehicle CPU **291** proceeds with the process to a step **4325** to determine whether the own vehicle **200** is braked. When the vehicle CPU **291** determines “Yes” at the step **4325**, the vehicle CPU **291** proceeds with the process to a step **4330** to wirelessly transmit the braking vibration request signal **S43** via the vehicle signal receiving-and-transmitting device **280**. Next, the vehicle CPU **291** proceeds with the process to the step **4340**.

(327) On the other hand, when the vehicle CPU **291** determines “No” at the step **4325**, the vehicle CPU **291** proceeds with the process to a step **4335** to wirelessly transmit the moving vibration request signal **S41** via the vehicle signal receiving-and-transmitting device **280**. Next, the vehicle CPU **291** proceeds with the process to the step **4340**.

(328) Further, when the vehicle CPU **291** determines “No” at the step **4310**, the vehicle CPU **291** proceeds with the process directly to the step **4340**.

(329) When the vehicle CPU **291** proceeds with the process to the step **4340**, the vehicle CPU **291** determines whether the shift state of the transmission apparatus **224** is changed from the forward driving state to the rearward driving state or from rearward driving state to the forward driving state. When the vehicle CPU **291** determines “Yes” at the step **4340**, the vehicle CPU **291** proceeds with the process to a step **4345** to wirelessly transmit the shift change vibration request signal **S42** via the vehicle signal receiving-and-transmitting device **280**. Next, the vehicle CPU **291** proceeds with the process to a step **4395** to terminate this routine once.

(330) On the other hand, when the vehicle CPU **291** determines “No” at the step **4340**, the vehicle CPU **291** proceeds with the process directly to the step **4395** to terminate this routine once.

(331) Further, when the vehicle CPU **291** determines “No” at the step **4305**, the vehicle CPU **291** proceeds with the process directly to the step **4395** to terminate this routine once.

(332) On the other hand, the terminal CPU **191** is configured or programmed to execute a routine shown in FIG. **44** with the predetermined calculation cycle when the vehicle CPU **291** is configured or programmed to execute the routine shown in FIG. **43**. Thus, at a predetermined timing, the terminal CPU **191** starts a process from a step **4400** of the routine shown in FIG. **44** and proceeds with the process to a step **4405** to determine whether the remote moving application software is executed.

(333) When the terminal CPU **191** determines “Yes” at the step **4405**, the terminal CPU **191** proceeds with the process to a step **4410** to determine whether to have received the obstacle vibration request signal **S44**. When the terminal CPU **191** determines “Yes” at the step **4410**, the terminal CPU **191** proceeds with the process to a step **4415** to execute the obstacle vibration process. Thereby, the operation terminal **100** is vibrated with the obstacle vibration pattern. Next, the terminal CPU **191** proceeds with the process to a step **4420**.

(334) On the other hand, when the terminal CPU **191** determines “No” at the step **4410**, the

terminal CPU **191** proceeds with the process directly to the step **4420**.

(335) When the terminal CPU **191** proceeds with the process to the step **4420**, the terminal CPU **191** determines whether to have received the braking vibration request signal **S43**. When the terminal CPU **191** determines “Yes” at the step **4420**, the terminal CPU **191** proceeds with the process to a step **4425** to execute the braking vibration process. Thereby, the operation terminal **100** is vibrated with the braking vibration pattern. Next, the terminal CPU **191** proceeds with the process to a step **4430**.

(336) On the other hand, when the terminal CPU **191** determines “No” at the step **4420**, the terminal CPU **191** proceeds with the process directly to the step **4430**.

(337) When the terminal CPU **191** proceeds with the process to the step **4430**, the terminal CPU **191** determines whether to have received the moving vibration request signal **S41**. When the terminal CPU **191** determines “Yes” at the step **4430**, the terminal CPU **191** proceeds with the process to a step **4435** to execute the moving vibration process. Thereby, the operation terminal **100** is vibrated with the moving vibration pattern. Next, the terminal CPU **191** proceeds with the process to a step **4440**.

(338) On the other hand, when the terminal CPU **191** determines “No” at the step **4430**, the terminal CPU **191** proceeds with the process directly to the step **4440**.

(339) When the terminal CPU **191** proceeds with the process to the step **4440**, the terminal CPU **191** determines whether to have received the shift change vibration request signal **S42**. When the terminal CPU **191** determines “Yes” at the step **4440**, the terminal CPU **191** proceeds with the process to a step **4445** to execute the shift change vibration process. Thereby, the operation terminal **100** is vibrated with the shift change vibration pattern. Next, the terminal CPU **191** proceeds with the process to a step **4495** to terminate this routine once.

(340) On the other hand, when the terminal CPU **191** determines “No” at the step **4440**, the terminal CPU **191** proceeds with the process directly to the step **4495** to terminate this routine once.

(341) Further, when the terminal CPU **191** determines “No” at the step **4405**, the terminal CPU **191** proceeds with the process directly to the step **4495** to terminate this routine once.

(342) The specific operations of the remote moving system **10** have been described.

(343) It should be noted that the invention is not limited to the aforementioned embodiments, and various modifications can be employed within the scope of the invention.

Claims

1. A non-transitory computer readable medium comprising a remote moving application software comprising machine readable instructions that, when executed by a processor, performs a remote moving control, wherein the remote moving application software is installed in a terminal control unit of an operation terminal to enable the terminal control unit to wirelessly communicate to a vehicle control unit of a vehicle to cause the vehicle control unit to execute the remote moving control to autonomously control braking and shifting of the vehicle so as to move, brake and change gears of the vehicle in response to user operations applied to the operation terminal, wherein the remote moving application software is programmed to control the terminal control unit to cause the operation terminal to (i) generate vibration or (ii) output informing sounds, depending on a state of controlling the vehicle when the remote moving control is executed, wherein when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is a state of changing a shift state of a transmission apparatus of the vehicle by autonomously shifting the vehicle, the remote moving application software is programmed to control the terminal control unit to cause the operation terminal to (i) generate the vibration having a vibration pattern different from the vibration pattern of the vibration generated by the operation terminal when the state of controlling the vehicle is a state other than the state of changing the shift state or (ii) output the

informing sounds having an output pattern different from the output pattern of the informing sounds output by the operation terminal when the state of controlling the vehicle is the state other than the state of changing the shift state.

2. The non-transitory computer readable medium as set forth in claim 1, wherein the remote moving application software is programmed to: when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is a state of moving the vehicle, control the terminal control unit to cause the operation terminal to (i) generate the vibration having a predetermined moving vibration pattern or (ii) output the informing sounds having a predetermined moving informing sound output pattern; and when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is the state of changing the shift state, control the terminal control unit to cause the operation terminal to (i) generate the vibration having a shift change vibration pattern different from the predetermined moving vibration pattern or (ii) output the informing sounds having a shift change informing sound output pattern different from the predetermined moving informing sound output pattern.

3. The non-transitory computer readable medium as set forth in claim 1, wherein the remote moving application software is programmed to: when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is a state of braking the vehicle, control the terminal control unit to cause the operation terminal to (i) generate the vibration having a predetermined braking vibration pattern or (ii) output the informing sounds having a predetermined braking informing sound output pattern; and when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is the state of changing the shift state, control the terminal control unit to cause the operation terminal to (i) generate the vibration having a shift changing vibration pattern different from the predetermined braking vibration pattern or (ii) output the informing sounds having a shift change informing sound output pattern different from the predetermined braking informing sound output pattern.

4. The non-transitory computer readable medium as set forth in claim 1, wherein the remote moving application software is programmed to: when (i) the remote moving control is executed, and (ii) an obstacle is detected around the vehicle, control the terminal control unit to cause the operation terminal to (i) generate the vibration having the vibration pattern different from the vibration pattern which depends on the state of controlling the vehicle or (ii) output the informing sounds having the output pattern different from the output pattern which depends on the state of controlling the vehicle.

5. A remote moving system which (i) causes a terminal control unit of an operation terminal and a vehicle control unit of a vehicle to wirelessly communicate to each other and (ii) causes the vehicle control unit to execute a remote moving control to autonomously control braking and shifting of the vehicle so as to move, brake and change gears of the vehicle in response to user operations applied to the operation terminal, wherein the remote moving system is configured to control the terminal control unit to cause the operation terminal to (i) generate vibration or (ii) output informing sounds, depending on a state of controlling the vehicle when the remote moving control is executed, wherein when (i) the remote moving control is executed, and (ii) the state of controlling the vehicle is a state of changing a shift state of a transmission apparatus of the vehicle by autonomously shifting the vehicle, the terminal control unit is configured to cause the operation terminal to (i) generate the vibration having a vibration pattern different from the vibration pattern of the vibration generated by the operation terminal when the state of controlling the vehicle is a state other than the state of changing the shift state or (ii) output the informing sounds having an output pattern different from the output pattern of the informing sounds output by the operation terminal when the state of controlling the vehicle is the state other than the state of changing the shift state.
